

# **2022 FALL**

# **Digital Image Processing**

## **1 Introduction**

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## Contents

This lecture will cover:

1. What is a digital image?
2. What is digital image processing?
3. Why do digital image processing?
4. History and state of the art examples of digital image processing
5. Key stages in digital image processing



***“One picture is worth more than ten thousand words”***

**Anonymous**

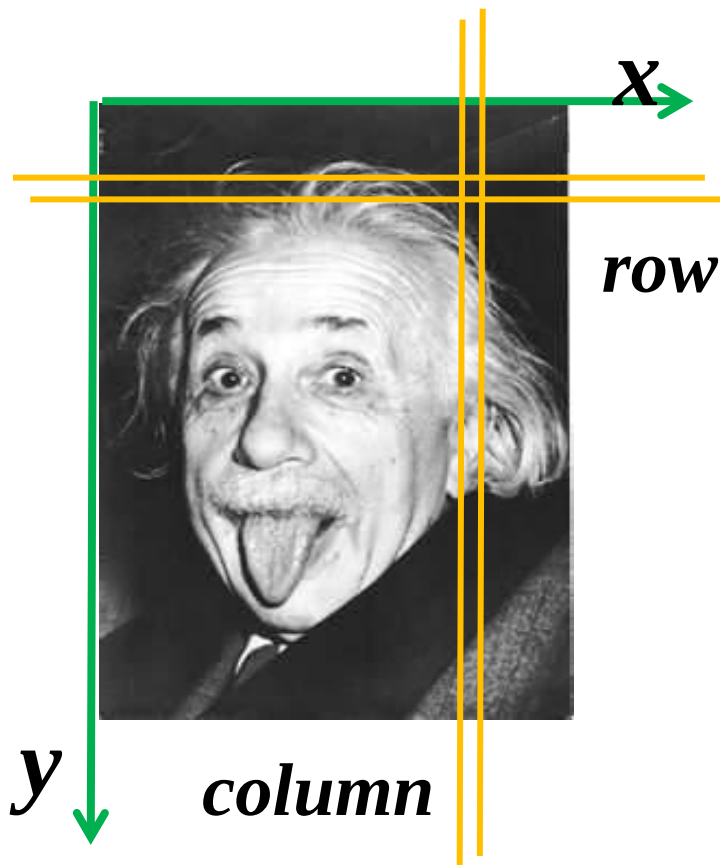
- Rich information from visual data
- Examples of images around us
  - Natural photographic images
  - Artistic and engineering drawings
  - Scientific images (satellite, medical, ocean etc.)
- Motion picture – video
  - Movies, TV programs, news
  - Family video
  - Surveillance and highway monitoring camera
  - Autonomous driving video



# *Digital image*



# 1. What is a Digital Image?

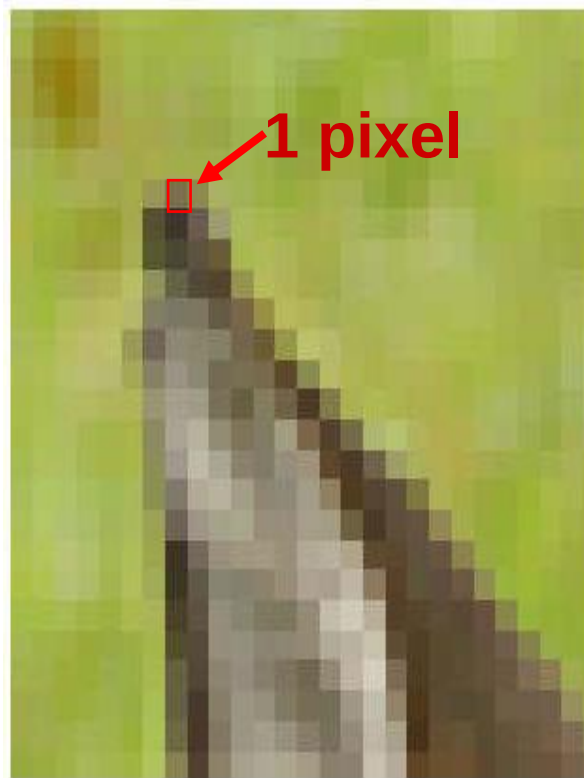
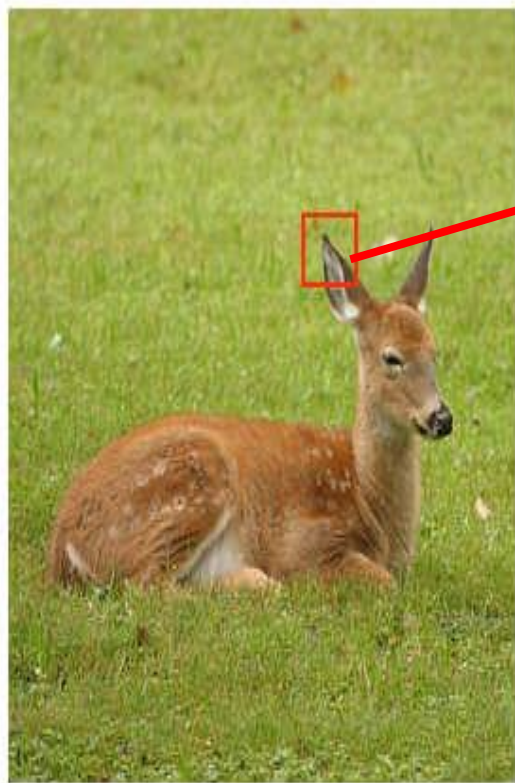


A **digital image**  $f(x, y)$  is a representation of a two-dimensional image as a finite set of digital values, called ***picture elements or pixels***.

- $x$  and  $y$  are spatial (plane) coordinates
- the amplitude of  $f$  at any pair of coordinates  $(x, y)$  is called the ***intensity or gray level*** of the image at that point.



# 1. What is a Digital Image?



***Pixel*** values typically represent:

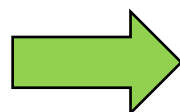
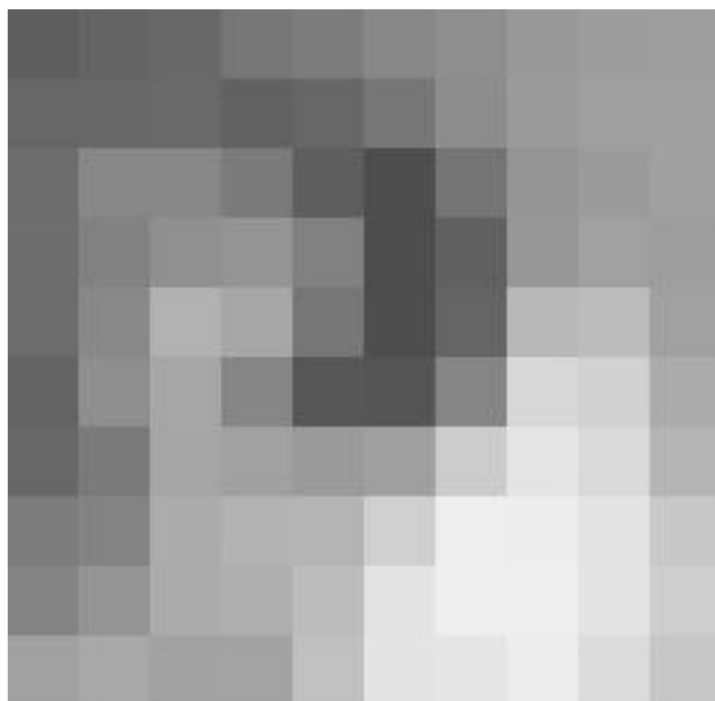
- gray levels
- Colors
- Heights
- opacities etc.

**Remember** *digitization implies that a digital image is an approximation of a real scene!*



# 1. What is a Digital Image?

Can be thought as a matrix of grey level/intensity values:



|     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 94  | 100 | 104 | 119 | 125 | 136 | 143 | 153 | 157 | 158 |
| 103 | 104 | 106 | 98  | 103 | 119 | 141 | 155 | 159 | 160 |
| 109 | 136 | 136 | 123 | 95  | 78  | 117 | 149 | 155 | 160 |
| 110 | 130 | 144 | 149 | 129 | 78  | 97  | 151 | 161 | 158 |
| 109 | 137 | 178 | 167 | 119 | 78  | 101 | 185 | 188 | 161 |
| 100 | 143 | 167 | 134 | 87  | 85  | 134 | 216 | 209 | 172 |
| 104 | 123 | 166 | 161 | 155 | 160 | 205 | 229 | 218 | 181 |
| 125 | 131 | 172 | 179 | 180 | 208 | 238 | 237 | 228 | 200 |
| 131 | 148 | 172 | 175 | 188 | 228 | 239 | 238 | 228 | 206 |
| 161 | 169 | 162 | 163 | 193 | 228 | 230 | 237 | 220 | 199 |



## 1. What is a Digital Image?

Common image formats include:

- 1 sample per point (B&W or Grayscale)
- 3 samples per point (Red, Green, and Blue)



For most of this course we will focus on grey-scale images





## 1. What is a Digital Image?

### ——Image type

- Color image



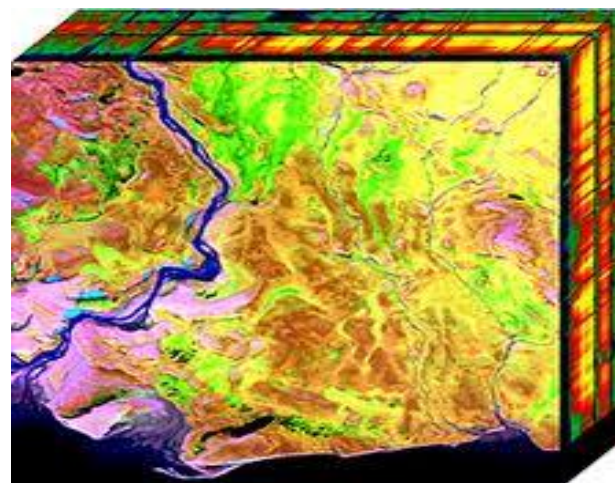
- Gray level image



- Infrared image

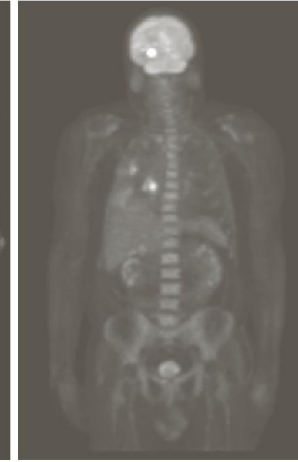
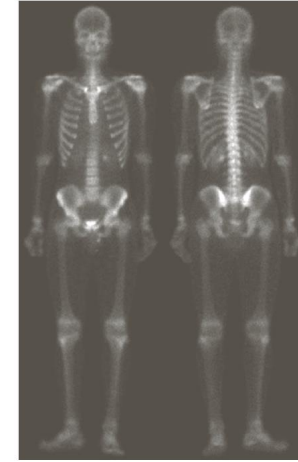
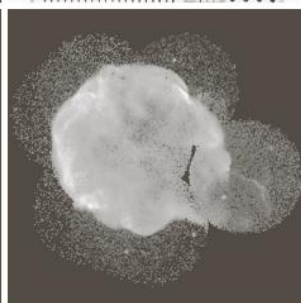
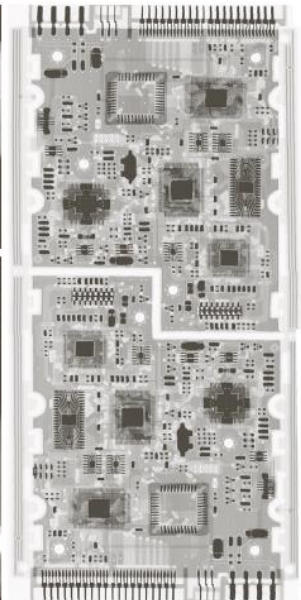


- Hyper-spectral image

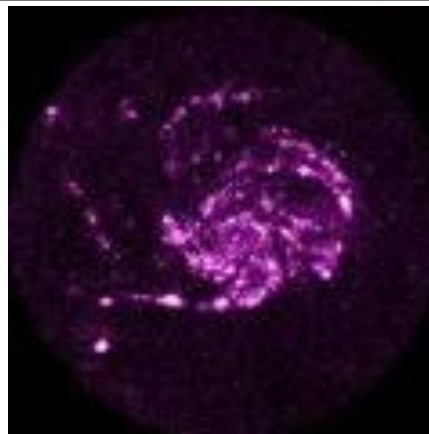


## 1. What is a Digital Image? —Image type

- X-ray images
- Ultrasound images
- Gamma-ray images



- Ultraviolet images
- MRI images



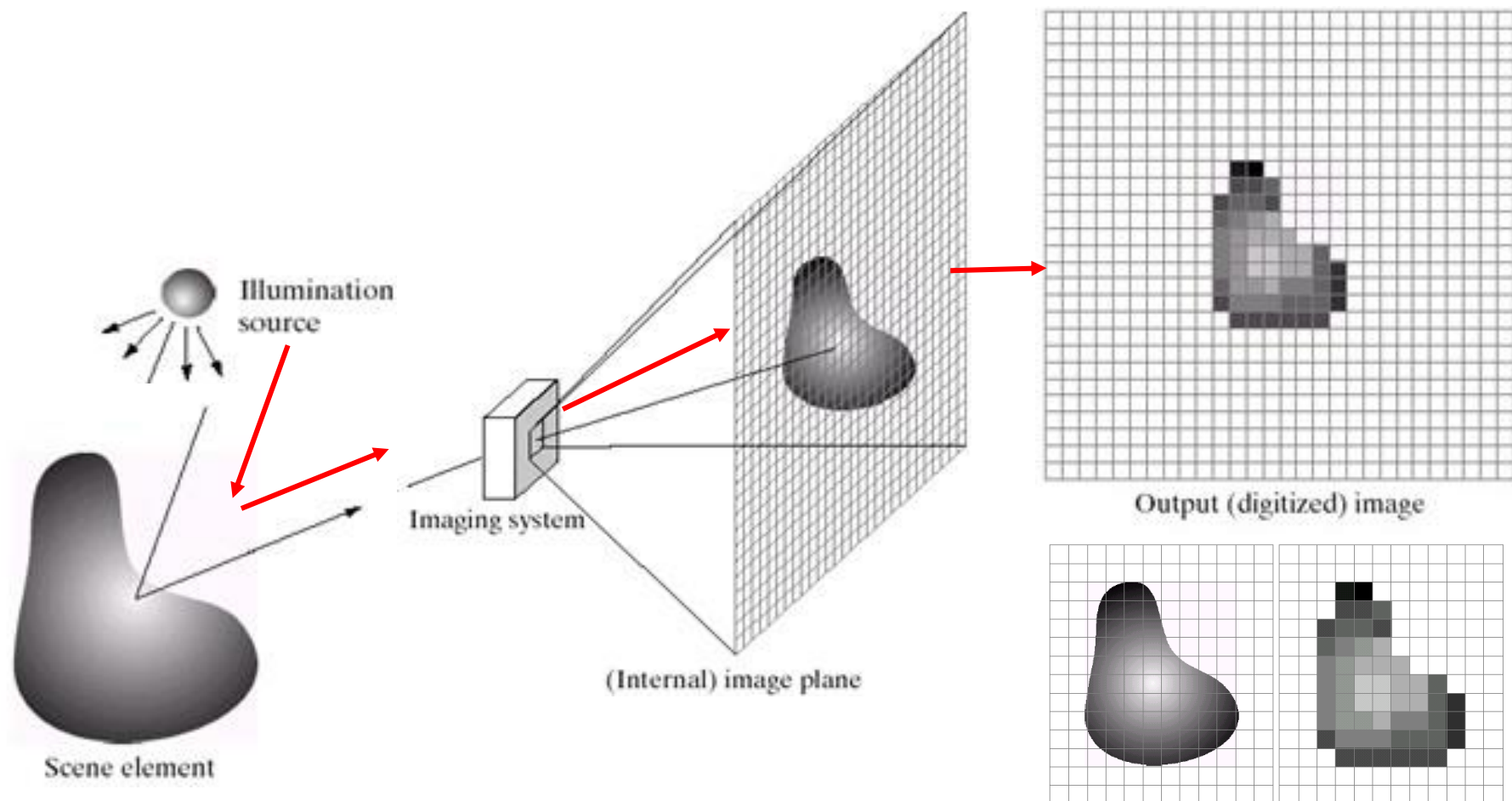
## 1. What is a Digital Image?

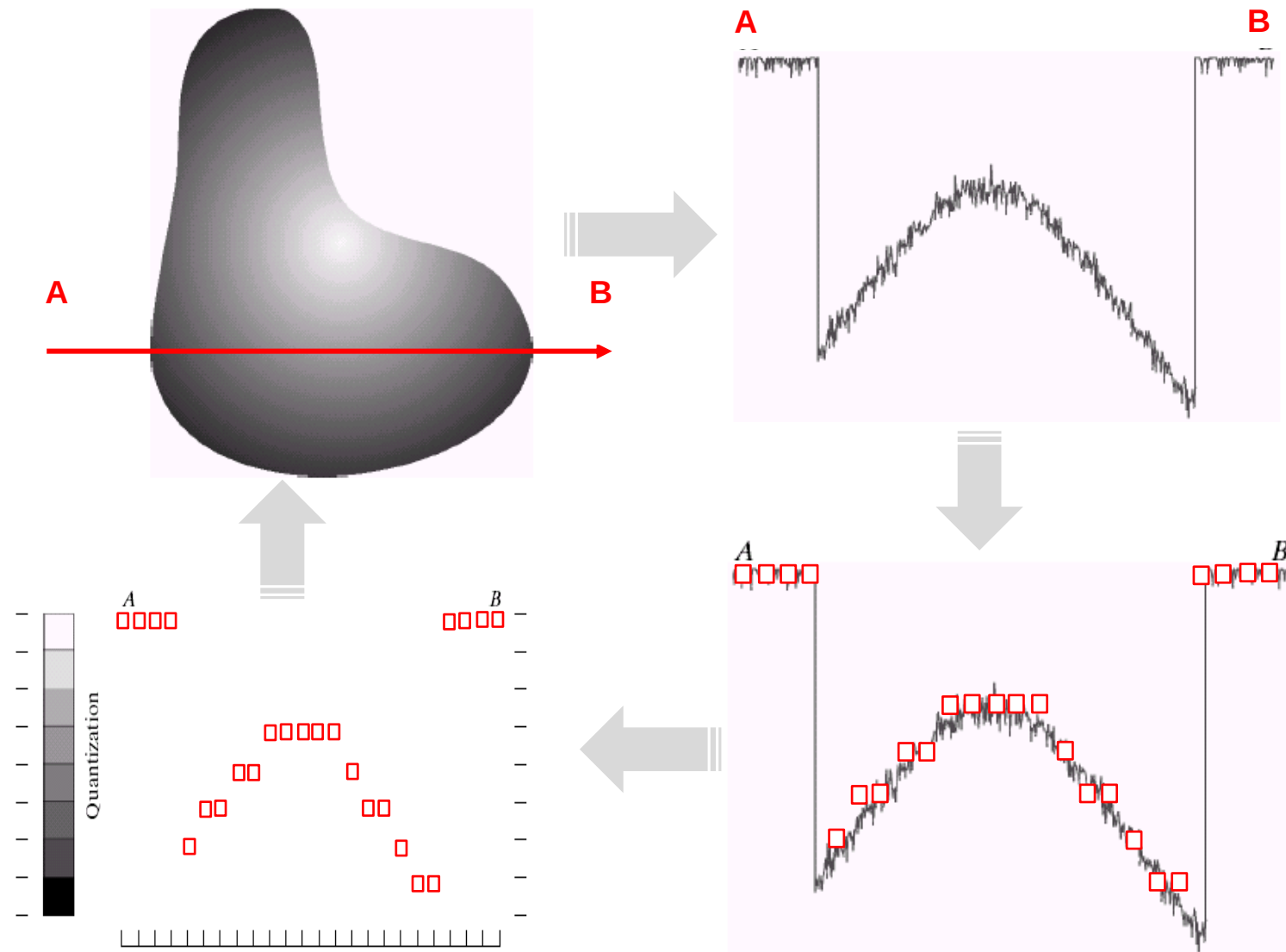
### Why Digital?

- “Exactness”
  - Perfect reproduction without degradation
  - Perfect duplication of processing result
- Convenient & powerful computer-aided processing
  - Can perform rather sophisticated processing through hardware or software
- Easy storage and transmission
  - 1 CD can store hundreds of family photos
  - Paperless transmission of high-quality images through network within seconds



## 1. What is a Digital Image?



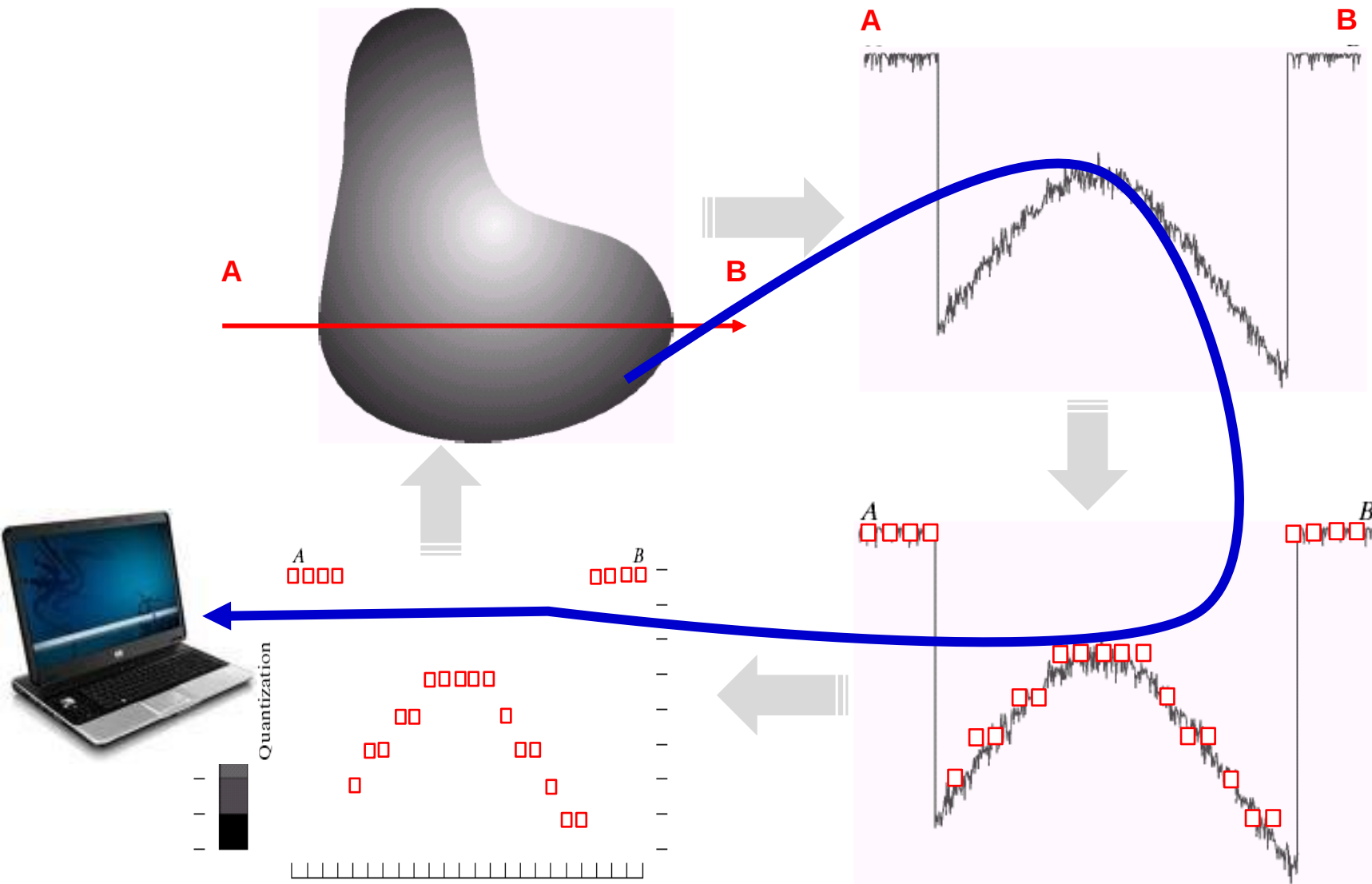


**Quantization-** Digitizing the amplitude values.

**Sampling-** Digitizing the coordinate values.







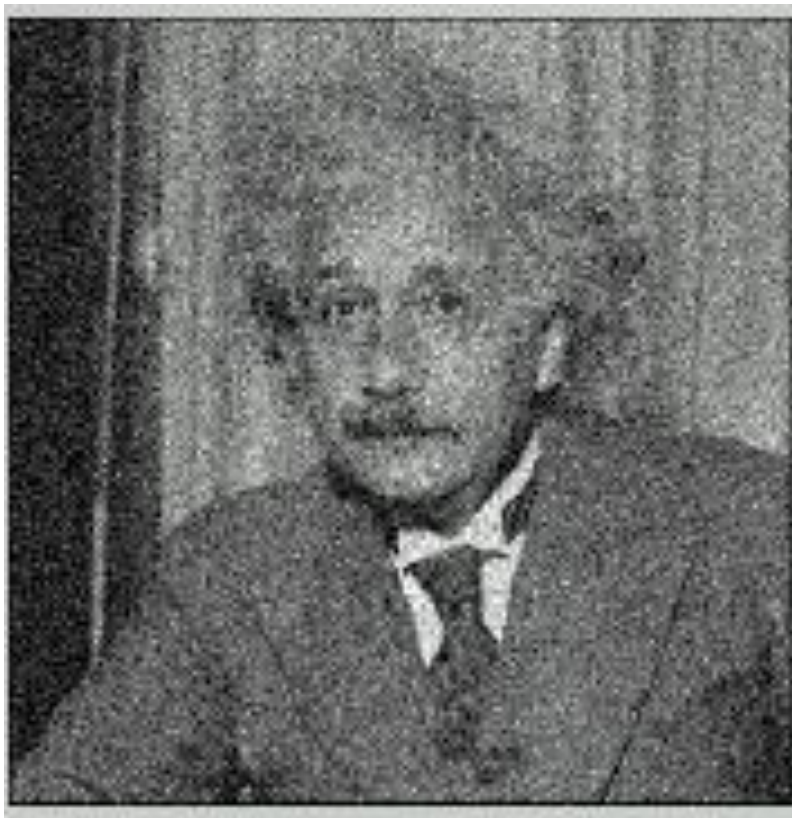
**Quantization-** Digitizing the amplitude values.

**Sampling-** Digitizing the coordinate values.

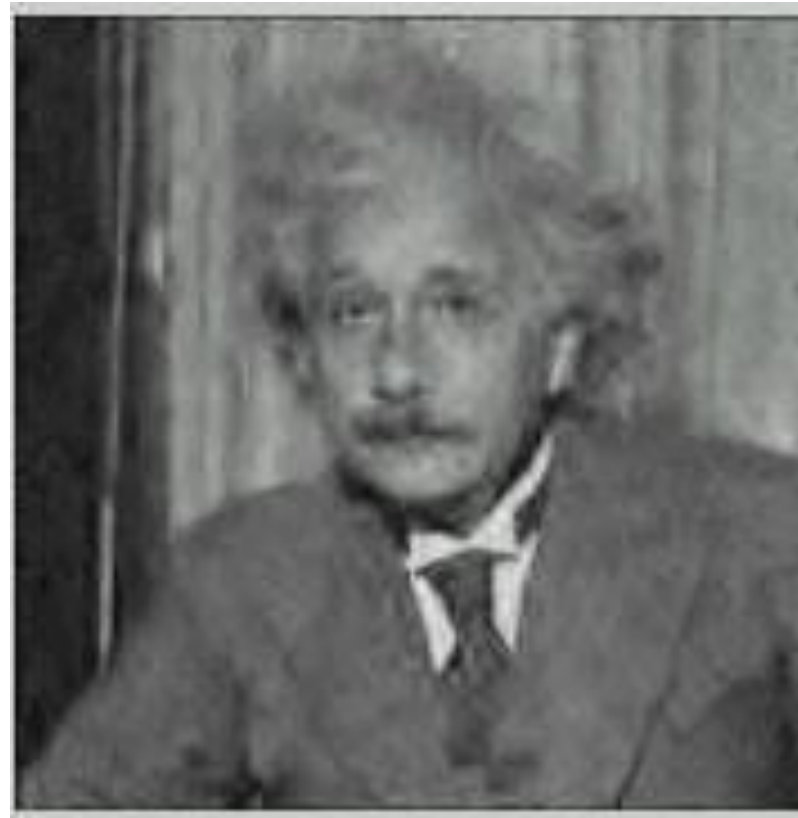


## 2. What's Digital Image Processing

- Process a digital image by means of a digital computer



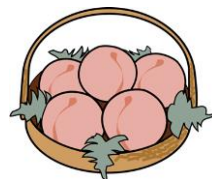
- Original



- Processed

**Enhancement!**





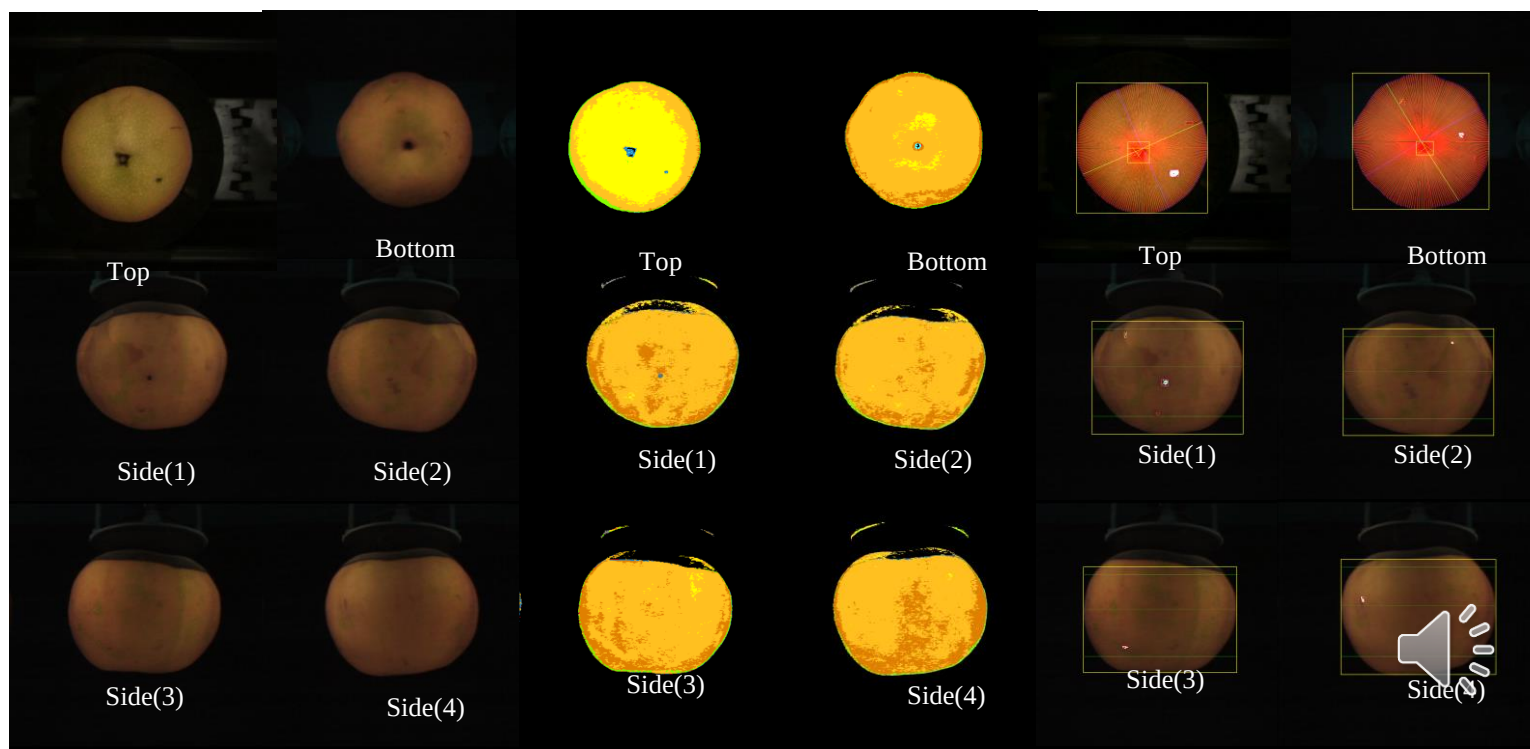
Pear  
Apple  
Peach  
Persimmon  
Tomato



## Analysis!

Image enhancement

Processed





## 2. What's Digital Image Processing

Image processing to computer vision

| Low Level                          | Middle Level                        | High Level                                    |
|------------------------------------|-------------------------------------|-----------------------------------------------|
| Noise removal,<br>image sharpening | Object recognition,<br>segmentation | Scene understanding,<br>autonomous navigation |

- Image improvement**

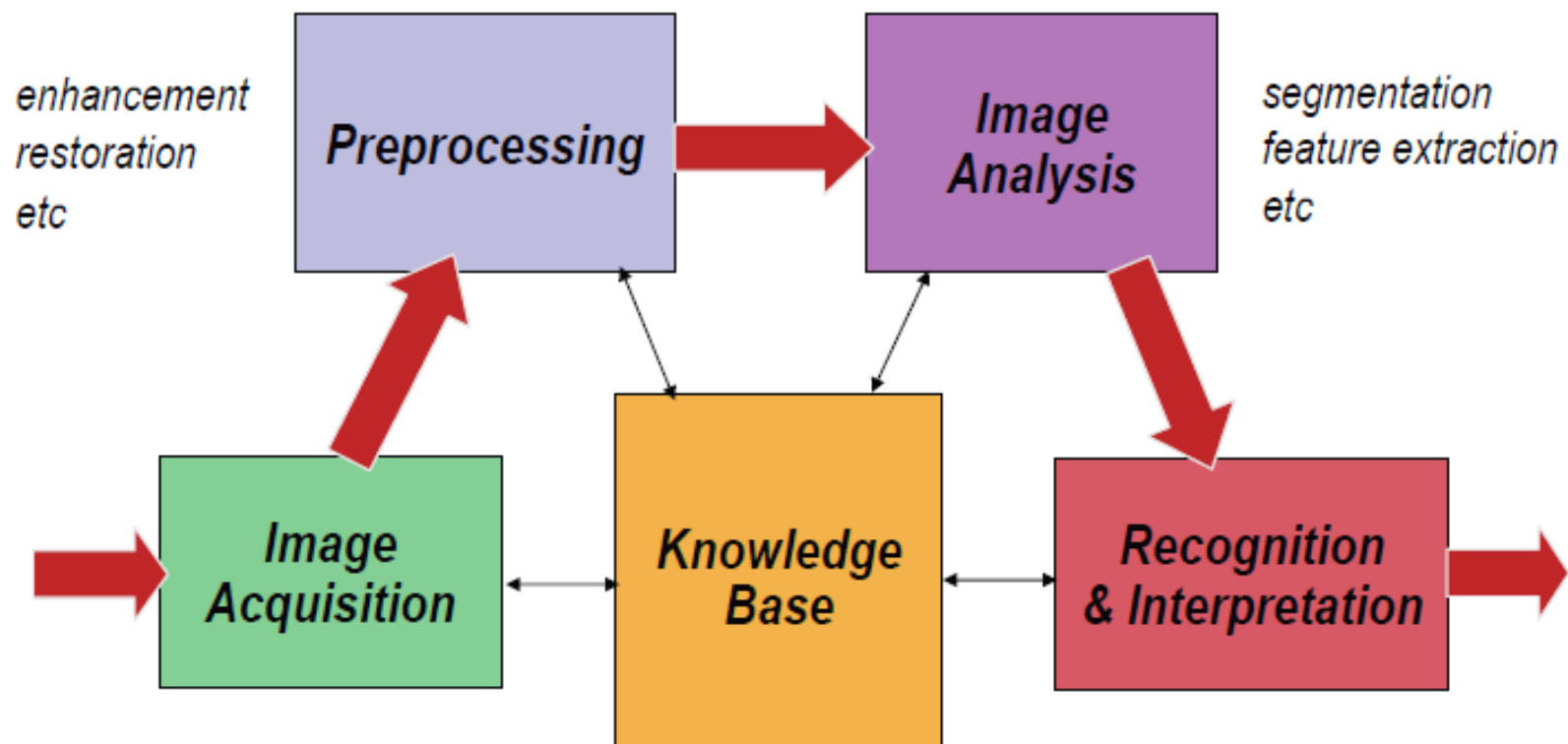
Improving the visual appearance of images to a human viewer

- Image analysis**

Preparing images for measurement of the features and structures present



## 2. What's Digital Image Processing



## 3. Why Do Digital Image Processing

### 1. For **Human Perception**

To make images more beautiful or understandable

### 2. **Automatic Perception** of Image

We call it Machine Vision, Computer Vision, Machine Perception, Computer Recognition

### 3. For **Storage and Transmission**

Smaller, faster, more effective

### 4. ***Image Processing for New Image Generation***

- Computer Graphics
- VR/AR



## Why do we process images?

### Acquire an image

- *Correct aperture and color balance*
- *Reconstruct image from projections*

### Prepare for display or printing

- *Adjust image size*
- *Color mapping, gamma-correction, halftoning*

### Facilitate picture storage and transmission

- *Efficiently store an image in a digital camera*
- *Send an image from space*

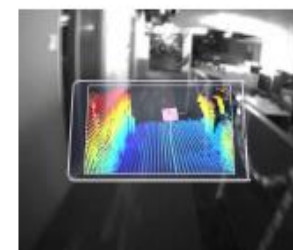
### Enhance and restore images

- *Touch up personal photos*
- *Color enhancement for security screening*

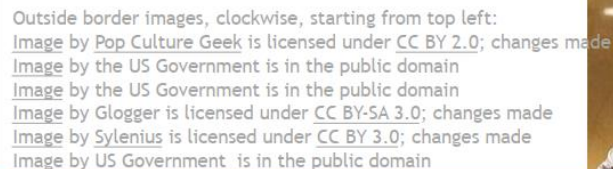
### Extract information from images

- *Read 2-d bar codes*
- *Character recognition*
- *Depth estimation*

Many more ... image processing is ubiquitous



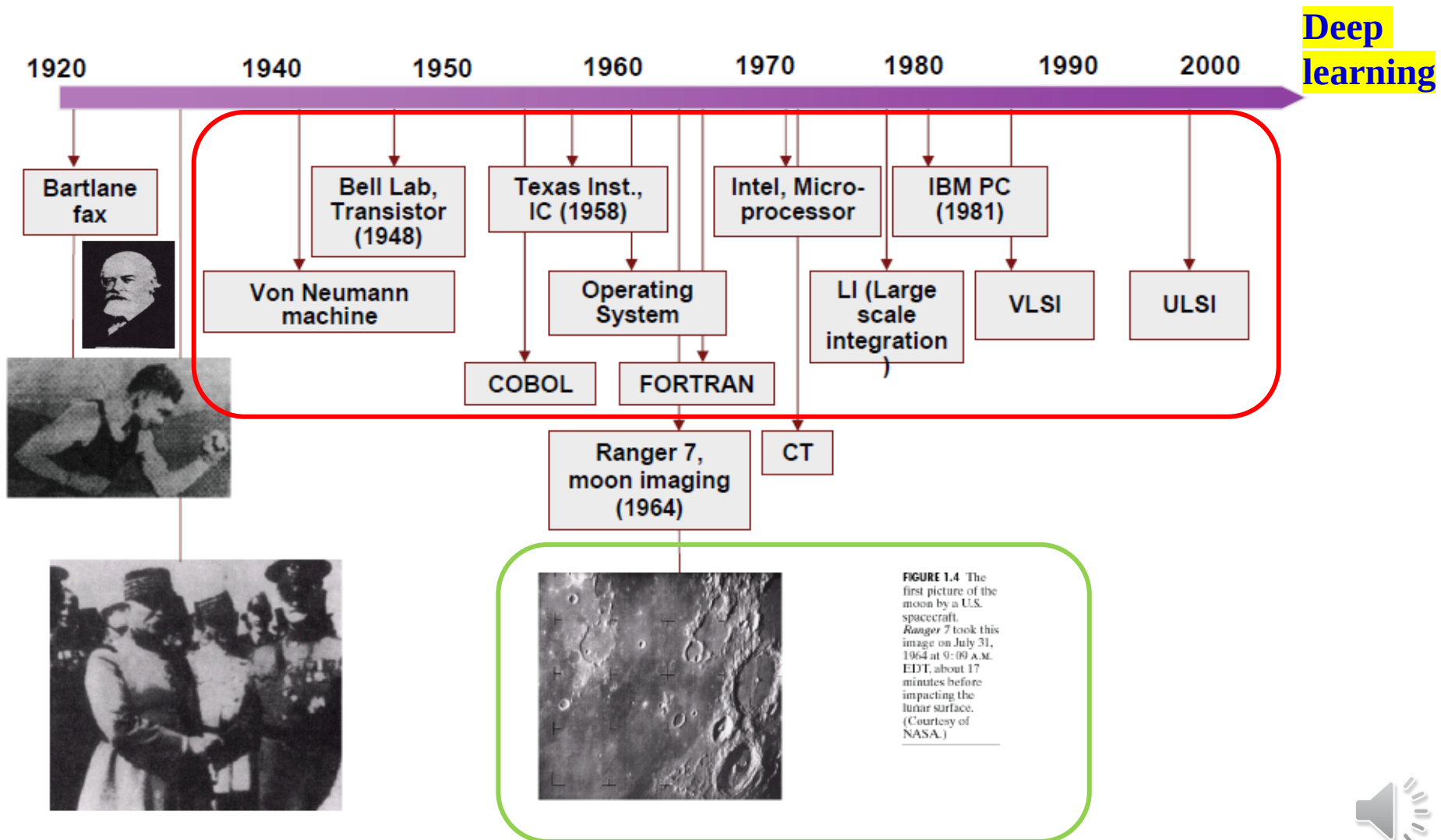




Inside four images, clockwise, starting from top left: Image is CC0 1.0 public domain; Image by Tucania is licensed under CC BY-SA 3.0; changes made by Image by Tucania are licensed under CC BY-SA 3.0; changes made by Oyundari Zorigbaatar is licensed under CC BY-SA 4.0.

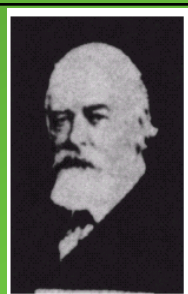


## 4. History of Digital Image Processing





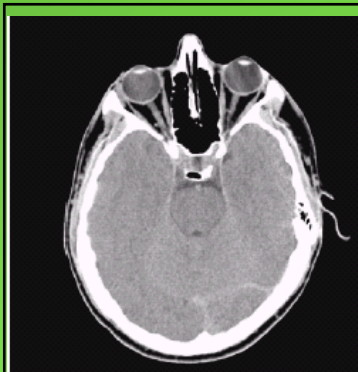
## 4. History of Digital Image Processing



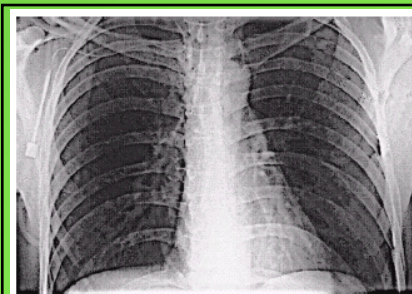
Improved  
digital image



A picture of the moon taken  
by the Ranger 7 probe  
minutes before landing



Typical head slice CAT image



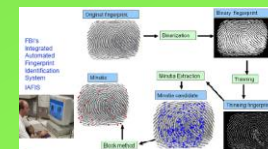
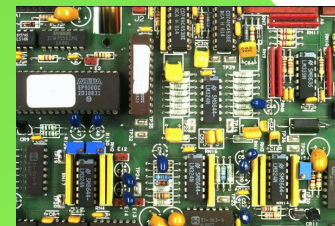
improve quality, remove noise etc

**1920s:**  
Early  
applications  
in the  
newspaper  
industry

**1960s:**  
Space race  
prompts  
digital image  
processing

**1970s:**  
Digital image  
processing  
begins to be  
used in medical  
applications

**1980s - Today:**  
For all kinds of  
tasks in all kinds of  
areas

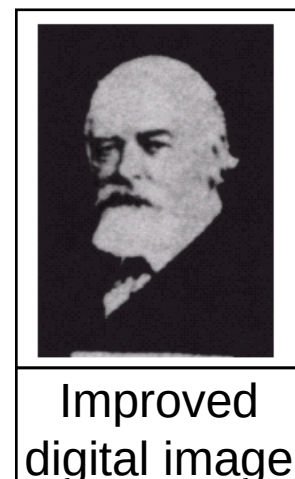


## 4. History of Digital Image Processing

**Early 1920s:** One of the first applications of digital imaging was in the newspaper industry



- The Bartlane cable picture transmission service
- Images were transferred by submarine cable between London and New York
- Pictures were coded for cable transfer and reconstructed at the receiving end on a telegraph printer



## 4. History of Digital Image Processing

**Mid to late 1920s:** Improvements to the Bartlane system resulted in higher quality images

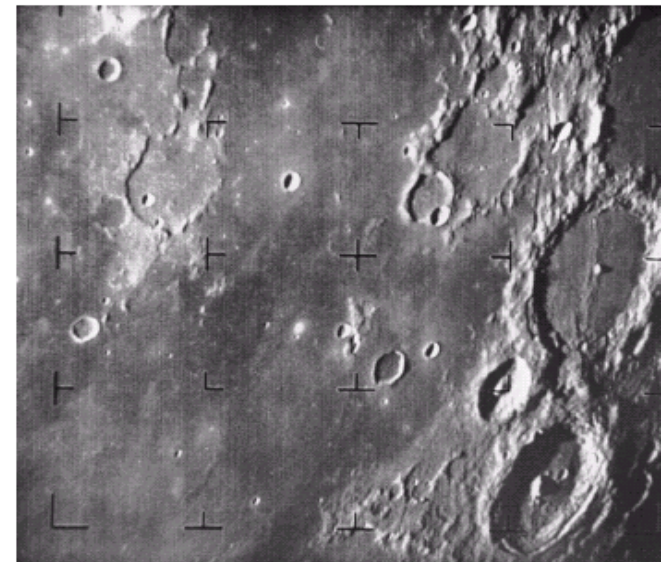
- New reproduction processes based on photographic techniques
- Increased number of tones in reproduced images



## 4. History of Digital Image Processing

**1960s:** Improvements in computing technology and the onset of the space race led to a surge of work in digital image processing

- **1964:** Computers used to improve the quality of images of the moon taken by the *Ranger 7* probe
- Such techniques were used in other space missions including the Apollo landings



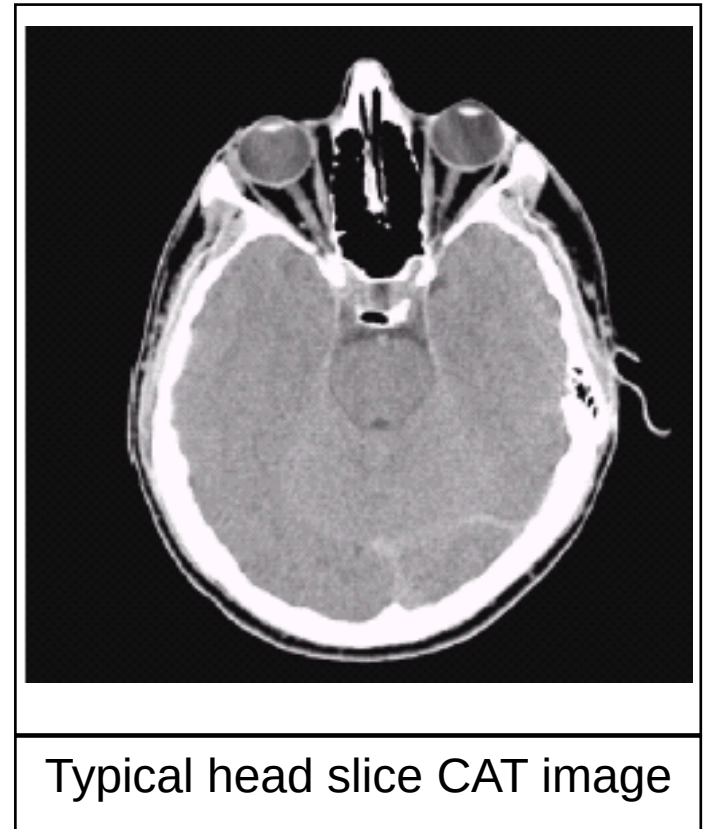
A picture of the moon taken by the Ranger 7 probe minutes before landing



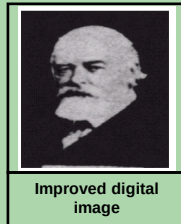
## 4. History of Digital Image Processing

**1970s:** Digital image processing begins to be used in **medical applications**

- **1979:** Sir Godfrey N. Hounsfield & Prof. Allan M. Cormack share the Nobel Prize in medicine for the invention of tomography, the technology behind Computerised Axial Tomography (CAT) scans

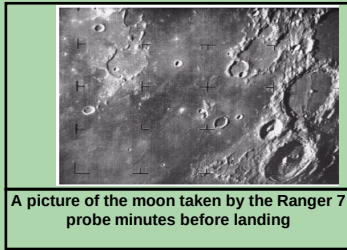






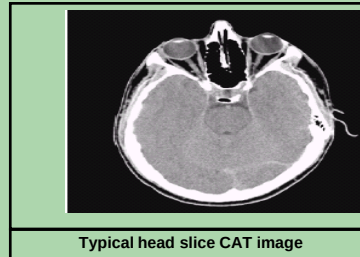
Improved digital image

1920s



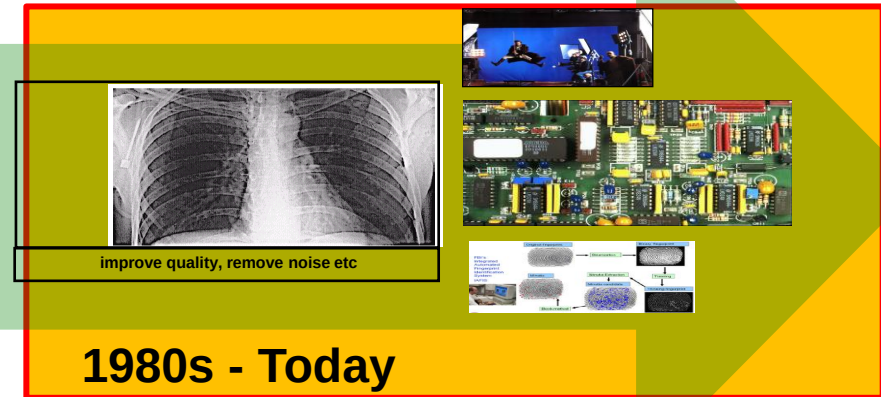
A picture of the moon taken by the Ranger 7 probe minutes before landing

1960s



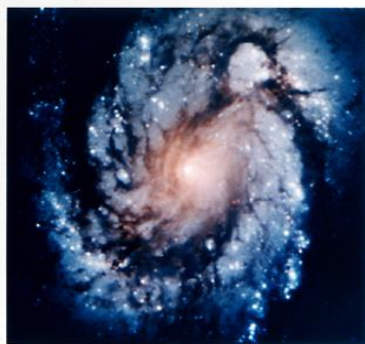
Typical head slice CAT image

1970s



1980s - Today

## (1) Space project



### Image enhancement

- 1990, Hubble telescope
- An incorrect mirror made many of Hubble's images useless
- Take images of very distant objects, send back to earth

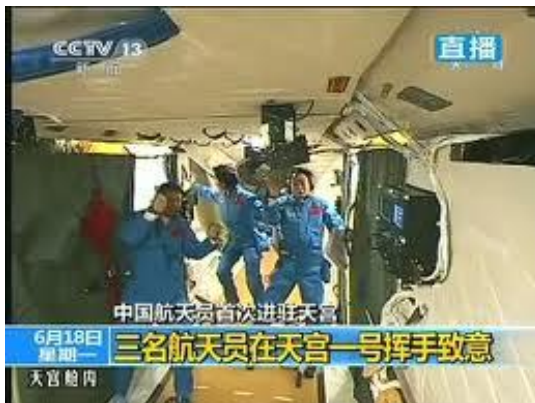
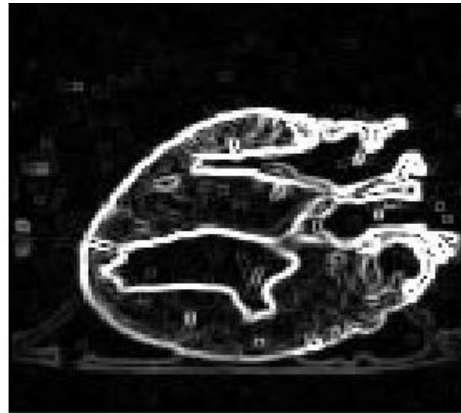
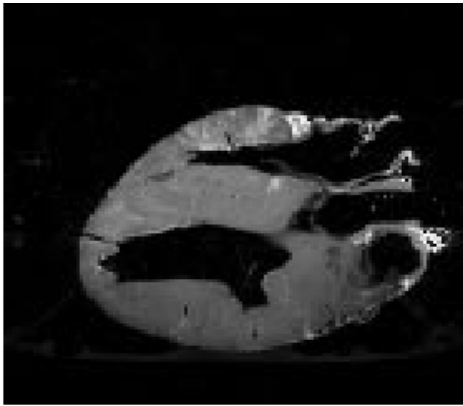


Image enhancement, image compression and coding

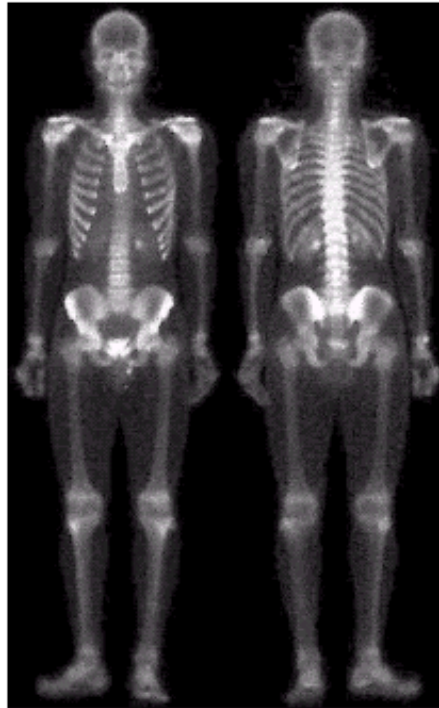
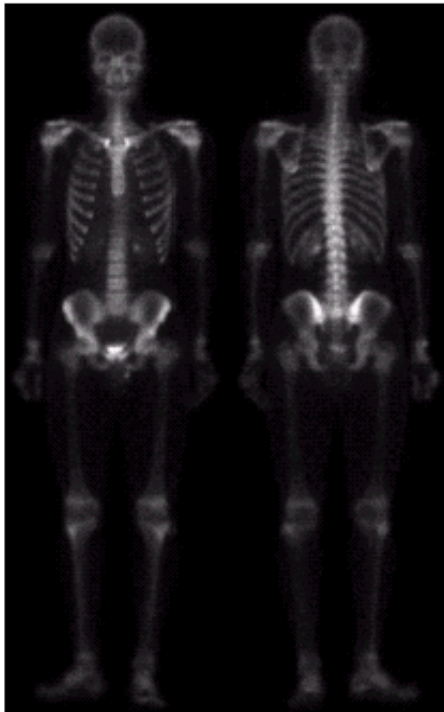


## (2) Medicine



Original MRI Image of a Dog Heart

**Edge detection**

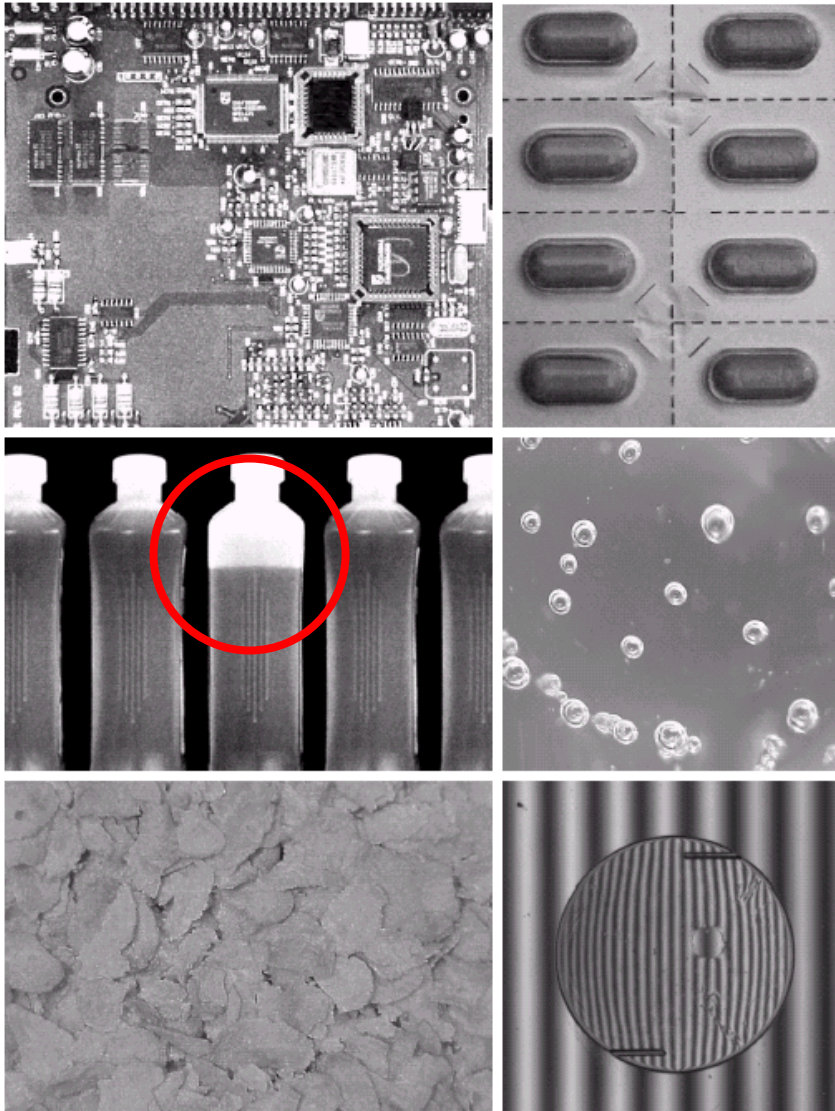


X-ray image of a human chest

**Image enhancement**



## (3) Industrial inspection



- Human operators are expensive, slow and unreliable
- Industrial vision systems take men's job.

**Object recognition**



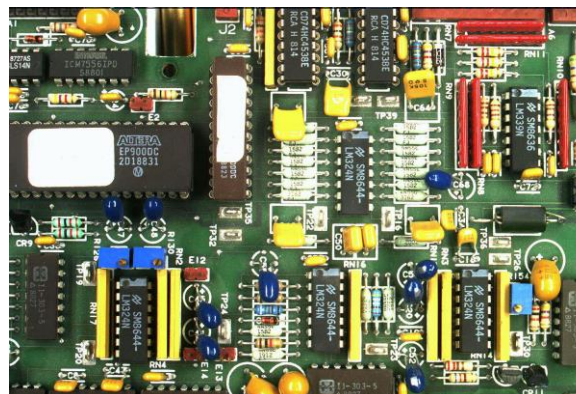
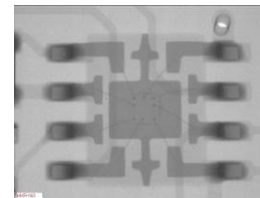
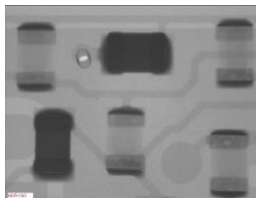


## (3) Industrial inspection

Examples: PCB Inspection

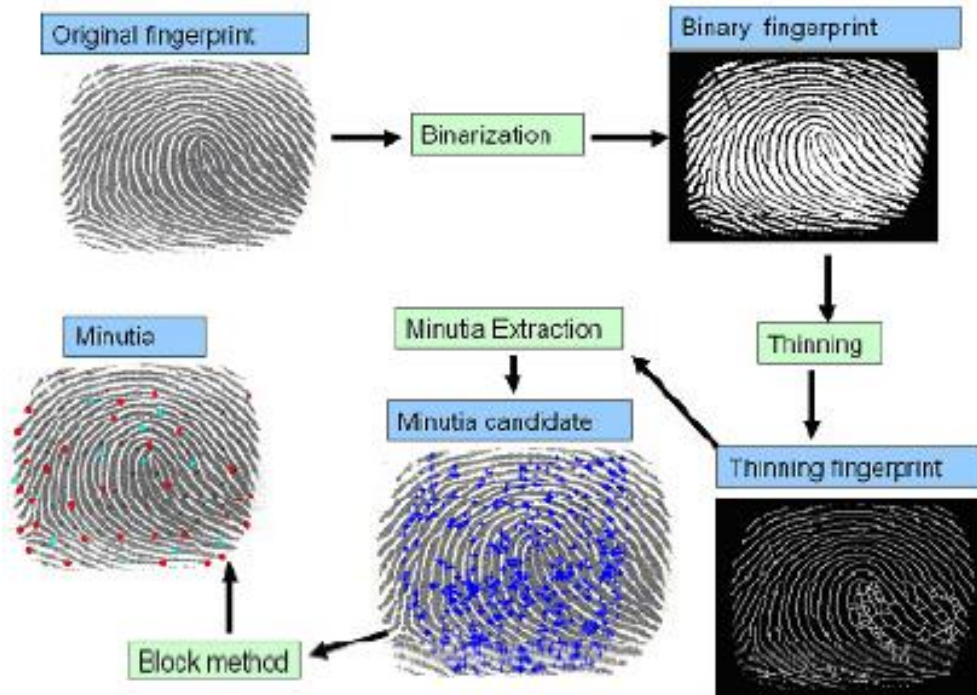
### Printed Circuit Board (PCB) inspection

- Machine inspection is used to determine that all components are present and that all solder joints are acceptable
- Both conventional imaging and x-ray imaging are used



## (4) Law enforcement

### Integrated Automated Fingerprint Identification System (IAFIS)



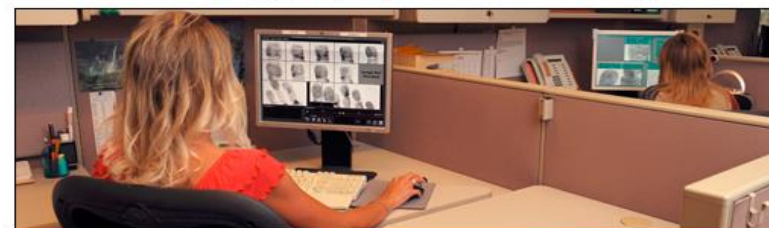
## Object matching and recognition

### Integrated Automated Fingerprint Identification System

[Home](#) • [About Us](#) • [CJIS](#) • [Fingerprints & Other Biometrics](#) • [IAFIS](#)

The Integrated Automated Fingerprint Identification System, or IAFIS, is a national fingerprint and criminal history system that responds to requests 24 hours a day, 365 days a year to help our local, state, and federal partners—and our own investigators—solve and prevent crime and catch criminals and terrorists. IAFIS provides automated fingerprint search capabilities, latent search capability, electronic image storage, and electronic exchange of fingerprints and responses.

**What is included in IAFIS:** Not only fingerprints, but corresponding criminal histories; mug shots; scars and tattoo photos; physical characteristics like height, weight, and hair and eye color; and aliases. The system also includes civil fingerprints, mostly of individuals who have served or are serving in the U.S. military or have been or are employed by the federal government. The fingerprints and criminal history information are submitted voluntarily by state, local, and federal law enforcement agencies.



The screenshot shows a license plate recognition software interface. It includes a video feed of a road with a blue truck and a black car. A table below the video lists vehicle information, including license plate numbers, directions, lanes, speeds, and status. To the right, there is a section for vehicle details, including the license plate number '粤C51625' and various fields for vehicle information.

| 车牌号码    | 行驶方向  | 车道 | 车速 | 车辆状态 | 通过时间                |
|---------|-------|----|----|------|---------------------|
| 粤CA1017 | 由东至西方 | 1  | 55 | 正常   | 2005-12-14 17:00:17 |
| 粤C84050 | 由东至西方 | 2  | 60 | 正常   | 2005-12-14 17:01:53 |
| 粤CA2927 | 由东至西方 | 2  | 57 | 正常   | 2005-12-14 17:01:58 |
| 粤A6M034 | 由东至西方 | 1  | 54 | 正常   | 2005-12-14 17:20:01 |
| 粤C51625 | 由东至西方 | 2  | 63 | 超速   | 2005-12-14 17:02:20 |

运行记录:

车辆信息:

疑似车牌: **粤C51625**

疑似车牌颜色:

车道: 2

行驶方向: 由东至西方

行驶速度(KM/H): 63

限制速度(KM/H): 60

车身长度: 4.5

车辆状态: 超速

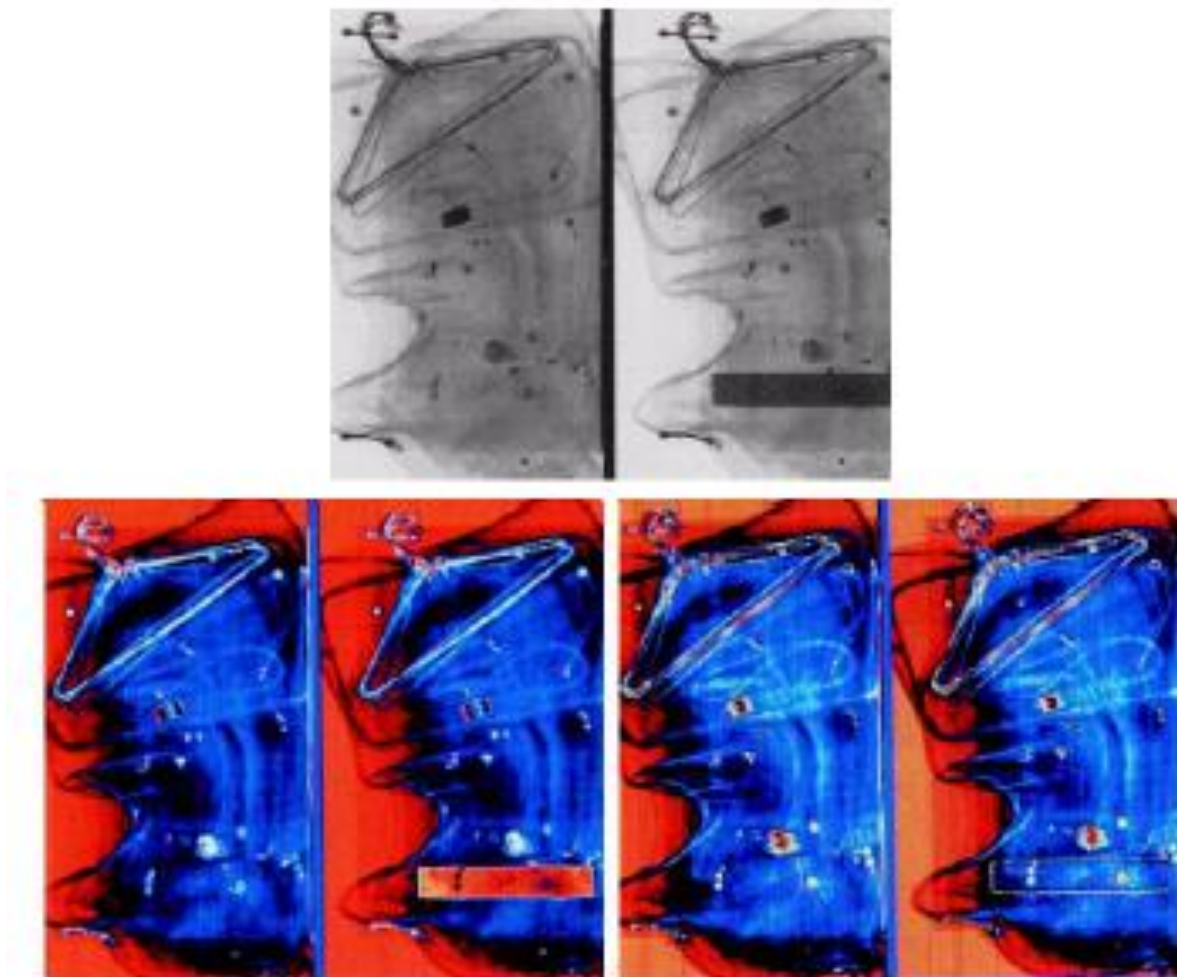
通过时间: 2005-12-14 17:02:20

## License plate recognition



## (4) Law enforcement

Examples: Pseudocolor enhancement

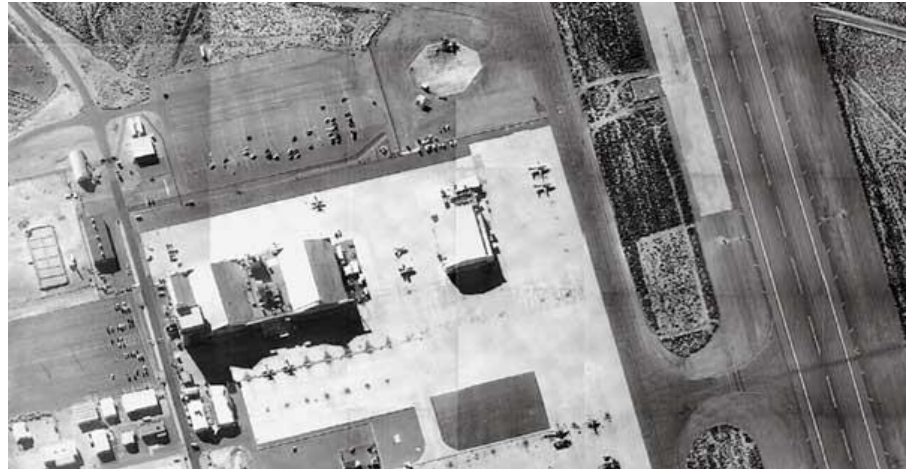


## (5) Military

Global Hawk, RQ4



Infrared image



Electrical optical image



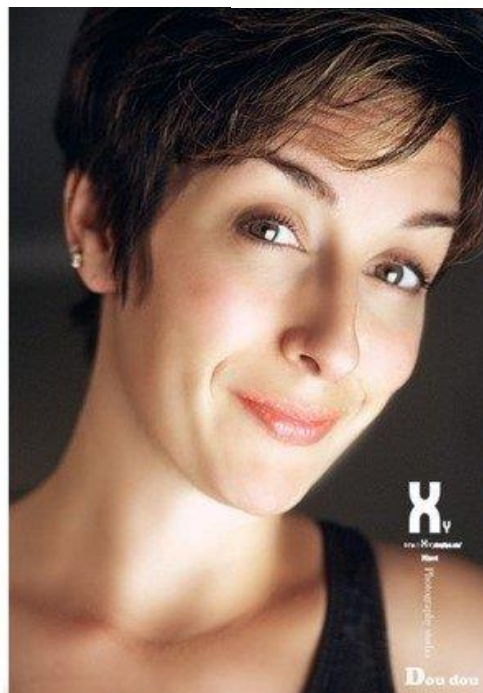
Radar image





## (6) Artistic effects

- Make images more visually appealing.
- Add special effects.
- Make composite images.







## (7) GIS

### Earthquake Analysis from Space

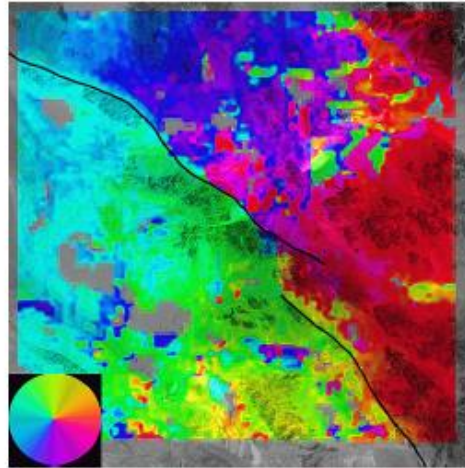
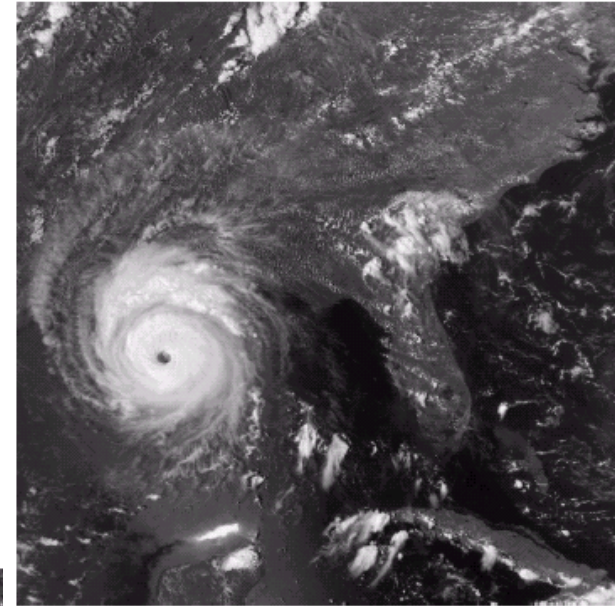


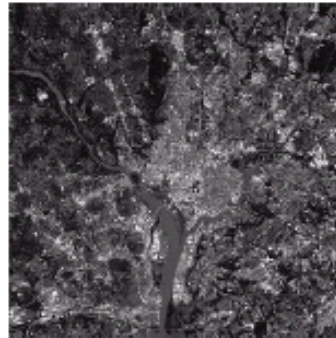
Image shows the ground displacement due to Landers earthquake in CA, 1992



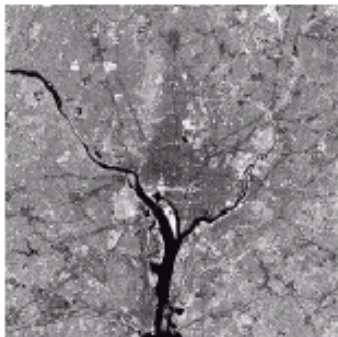
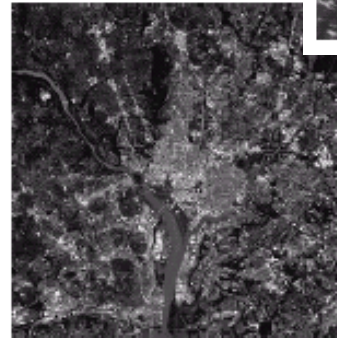
1



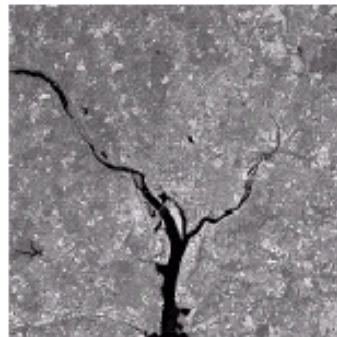
2



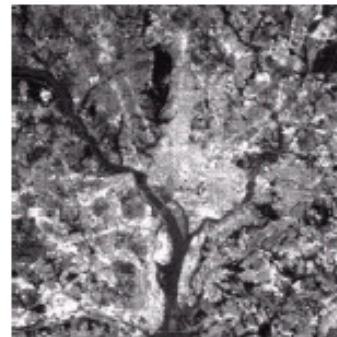
3



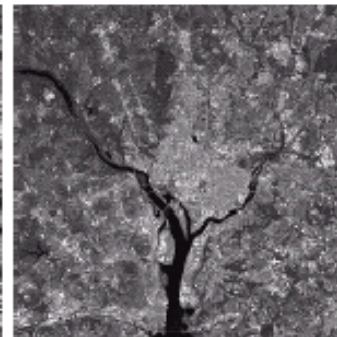
4



5



6



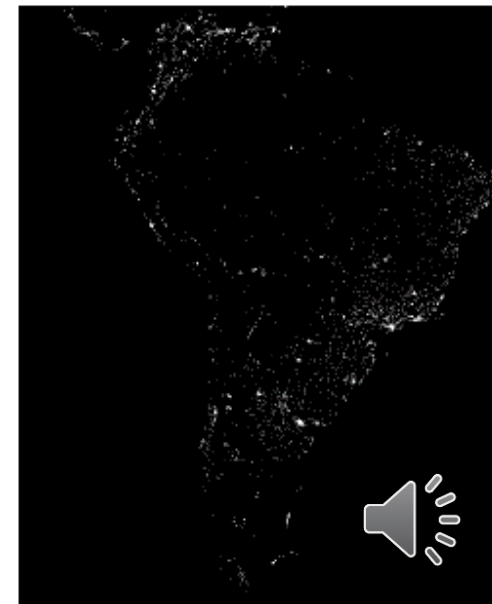
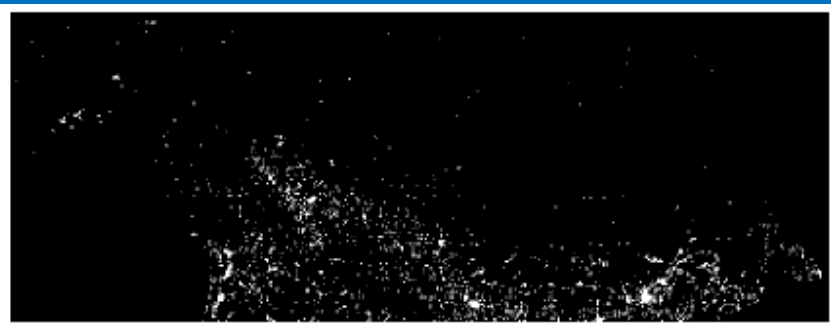
7



## (7) GIS

### *Night-Time Lights of the World data set*

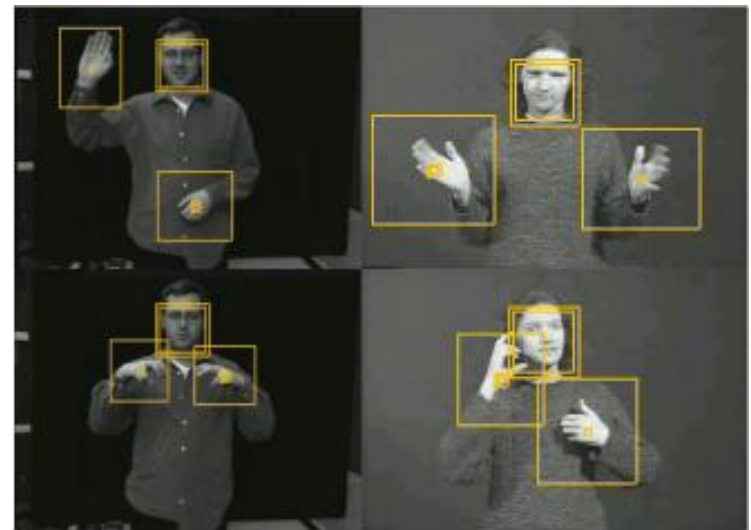
- Global inventory of human settlement
- Not hard to imagine the kind of analysis that might be done using this data



## (8) HIC and handwriting

Try to make human computer interfaces more natural

- Face recognition
- Gesture recognition
- Handwriting recognition



## (8) HIC and handwriting

0 1 2 3 4 5 6 7 8 9

0 1 2 3 4 5 6 7 8 9

0 1 2 3 4 5 6 7 8 9

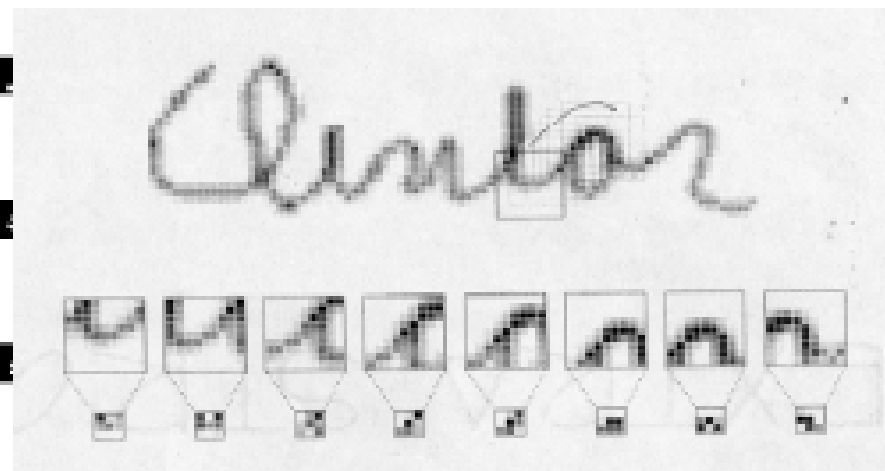
(a)

0 1 2 3 4

0 1 2 3 4

0 1 2 3 4

(b)



0 1 2 3 4 5 6 7 8 9

0 1 2 3 4 5 6 7 8 9

0 1 2 3 4 5 6 7 8 9

(c)

0 1 2 3 4 5 6 7 8 9

0 1 2 3 4 5 6 7 8 9

0 1 2 3 4 5 6 7 8 9

(d)





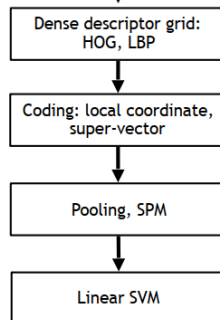
## (9) Deep learning-base DIP

- Convolutional Neural Networks (CNN) have become an important tool

### IMAGENET Large Scale Visual Recognition Challenge

#### Year 2010

NEC-UIUC

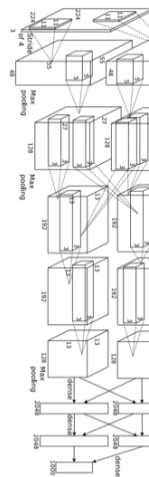


[Lin CVPR 2011]

Lion image by Swissfrog is  
licensed under CC BY 3.0

#### Year 2012

SuperVision

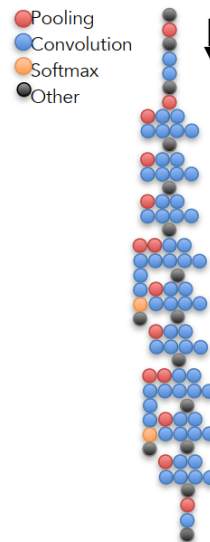


[Krizhevsky NIPS 2012]

Figure copyright Alex Krizhevsky, Ilya  
Sutskever, and Geoffrey Hinton, 2012.  
Reproduced with permission.

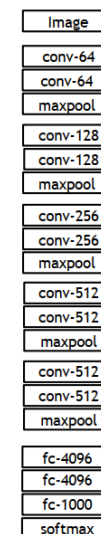
#### Year 2014

GoogLeNet



[Szegedy arxiv 2014]

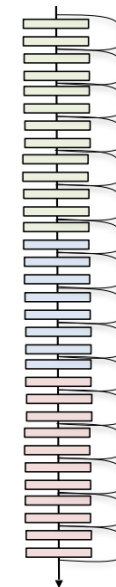
VGG



[Simonyan arxiv 2014]

#### Year 2015

MSRA



[He ICCV 2015]

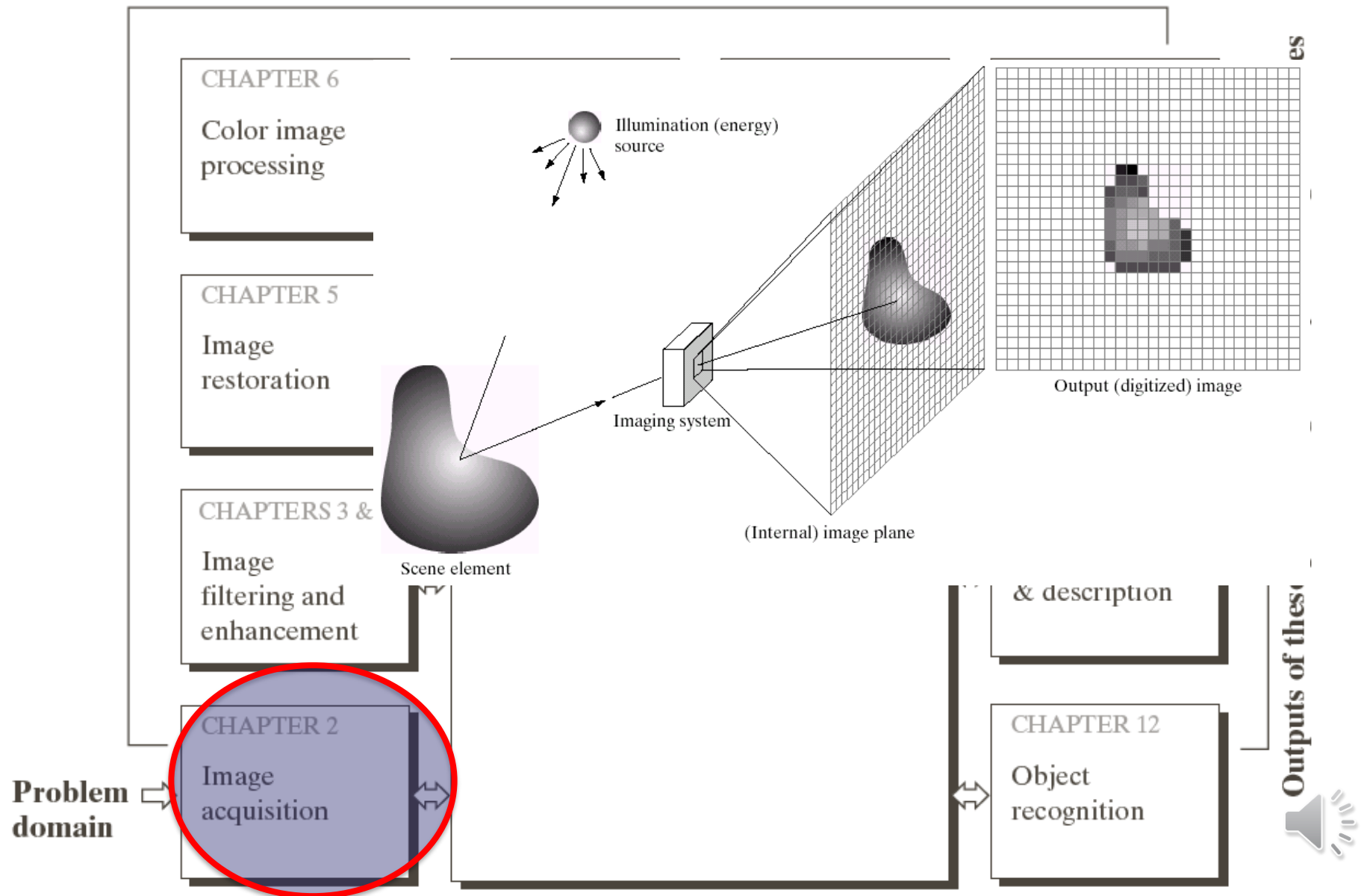


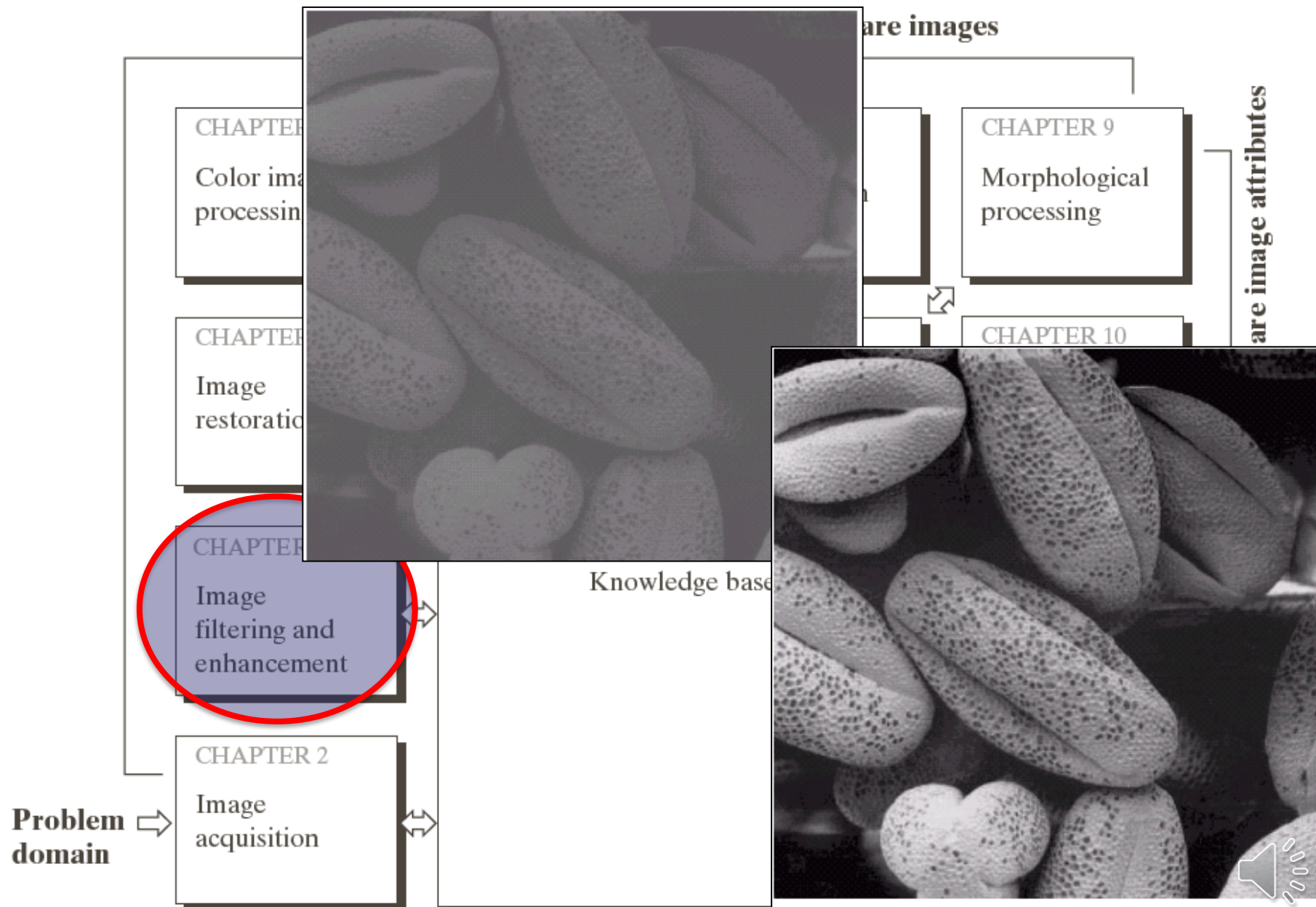
-  DigitallImageProcessing1-Introduction.pptx
-  DigitallImageProcessing2-ImageProcessingFundamentals.ppt
-  DigitallImageProcessing3-ImageEnhancement-point.ppt
-  DigitallImageProcessing4-ImageEnhancement-spatial filter.ppt
-  DigitallImageProcessing5-ImageEnhancement-frequency.ppt
-  DigitallImageProcessing6-Edge detection.ppt
-  DigitallImageProcessing7-Segmentation.ppt
-  DigitallImageProcessing8-Morphological.ppt
-  DigitallImageProcessing9-colorprocessing.ppt
-  DigitallImageProcessing10-representation.ppt
-  DigitallImageProcessing11-Keypoint Detection.pptx
-  DigitallImageProcessing12-Scale-Space Image Processing.pptx
-  DigitallImageProcessing13-Feature-based Methods for Image Matching.pptx
-  DigitallImageProcessing14-Panoramic Imaging.pptx
-  DigitallImageProcessing15-CNN for DIP.ppt
-  DigitallImageProcessing16-review.ppt



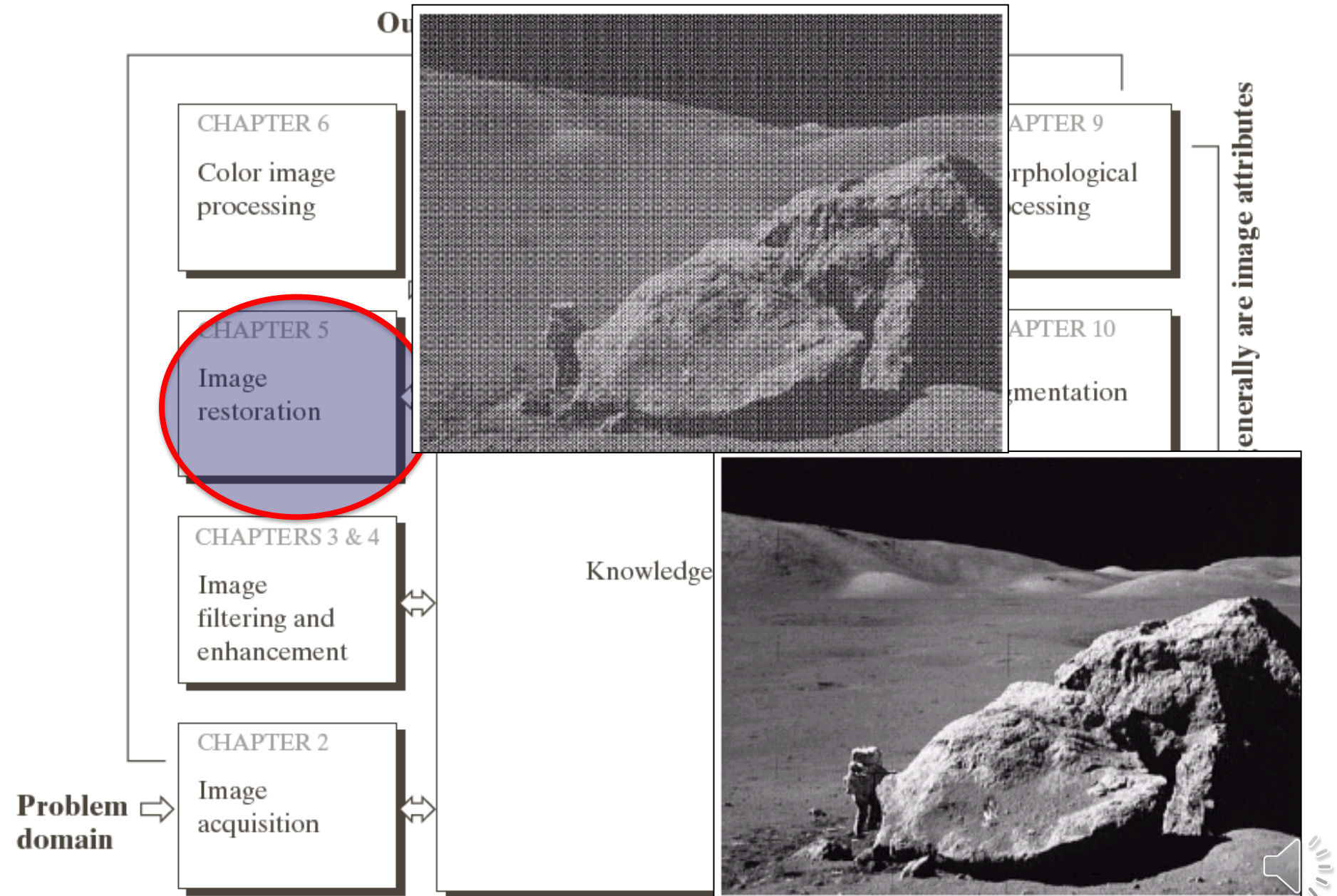
## 5. Key stages in digital image processing

Outputs of these processes generally are images









Outputs of these processes generally are images

CHAPTER 6

Color image processing

CHAPTER 7

Wavelets and multiresolution processing

CHAPTER 8

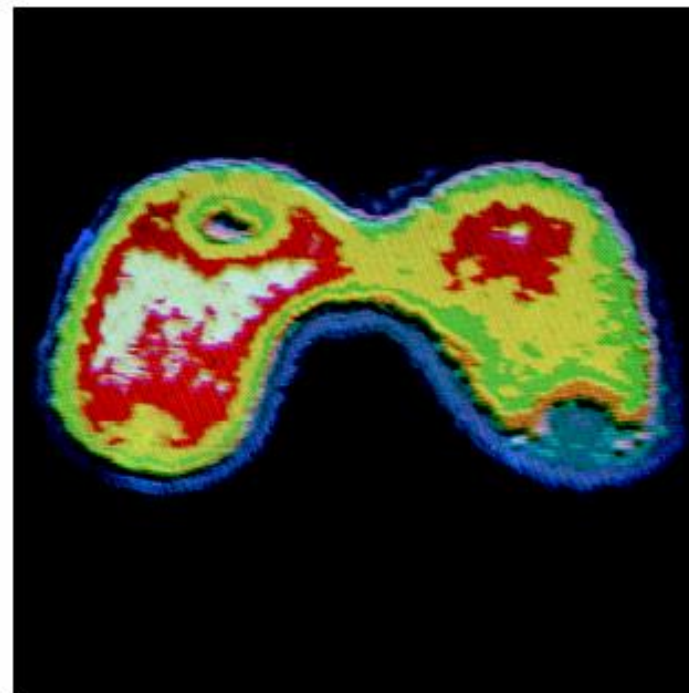
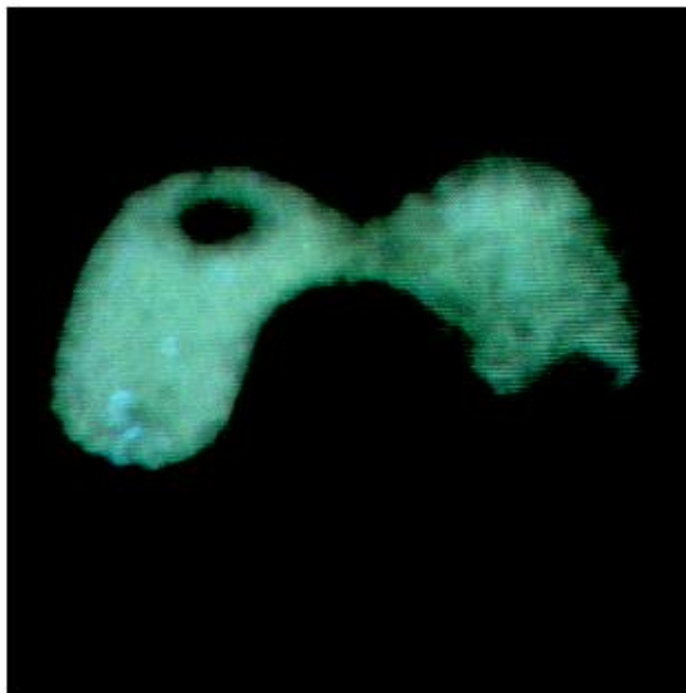
Compression

CHAPTER 9

Morphological processing

CHAPTER 5

CHAPTER 10

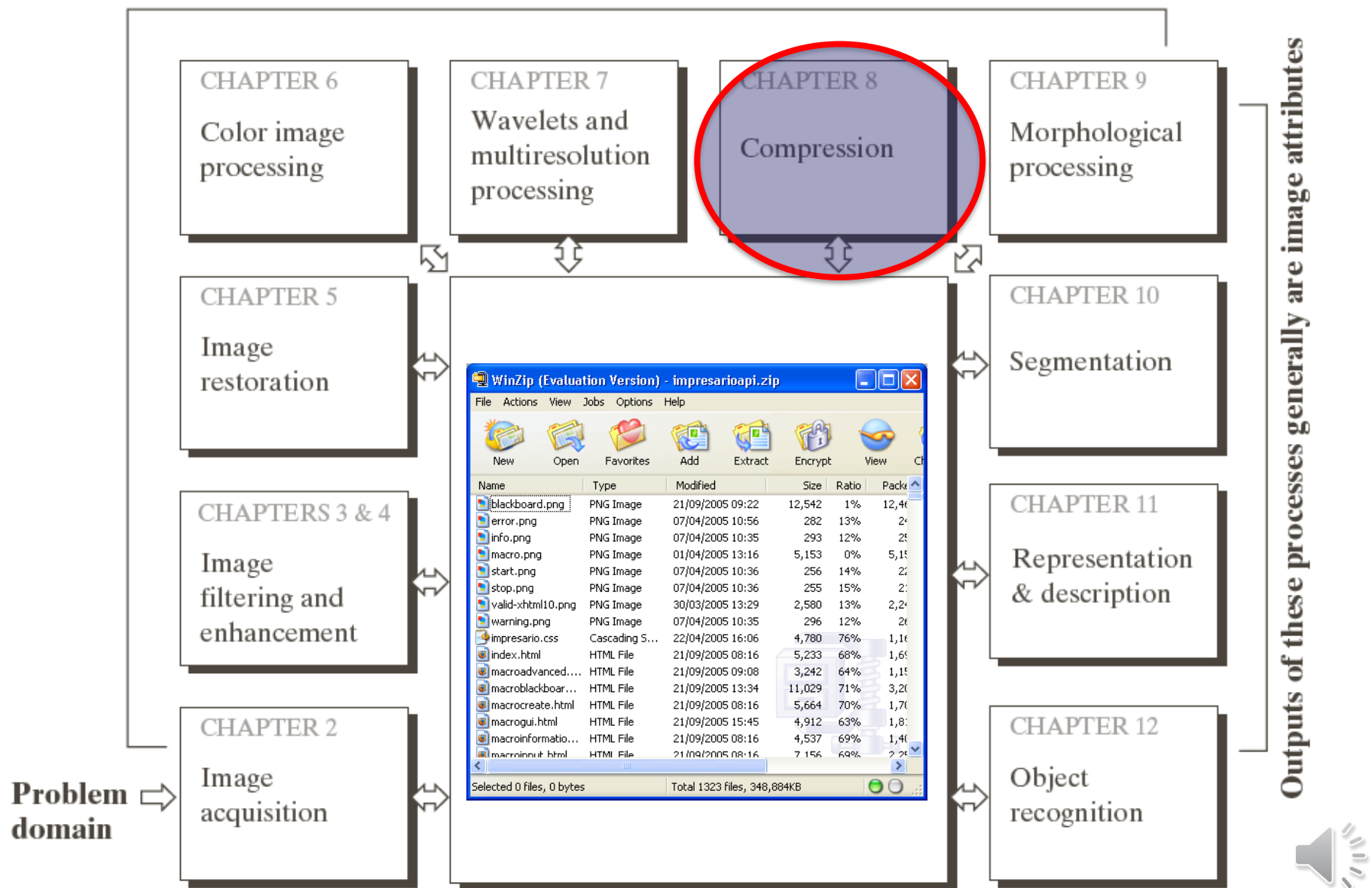


Problem domain →

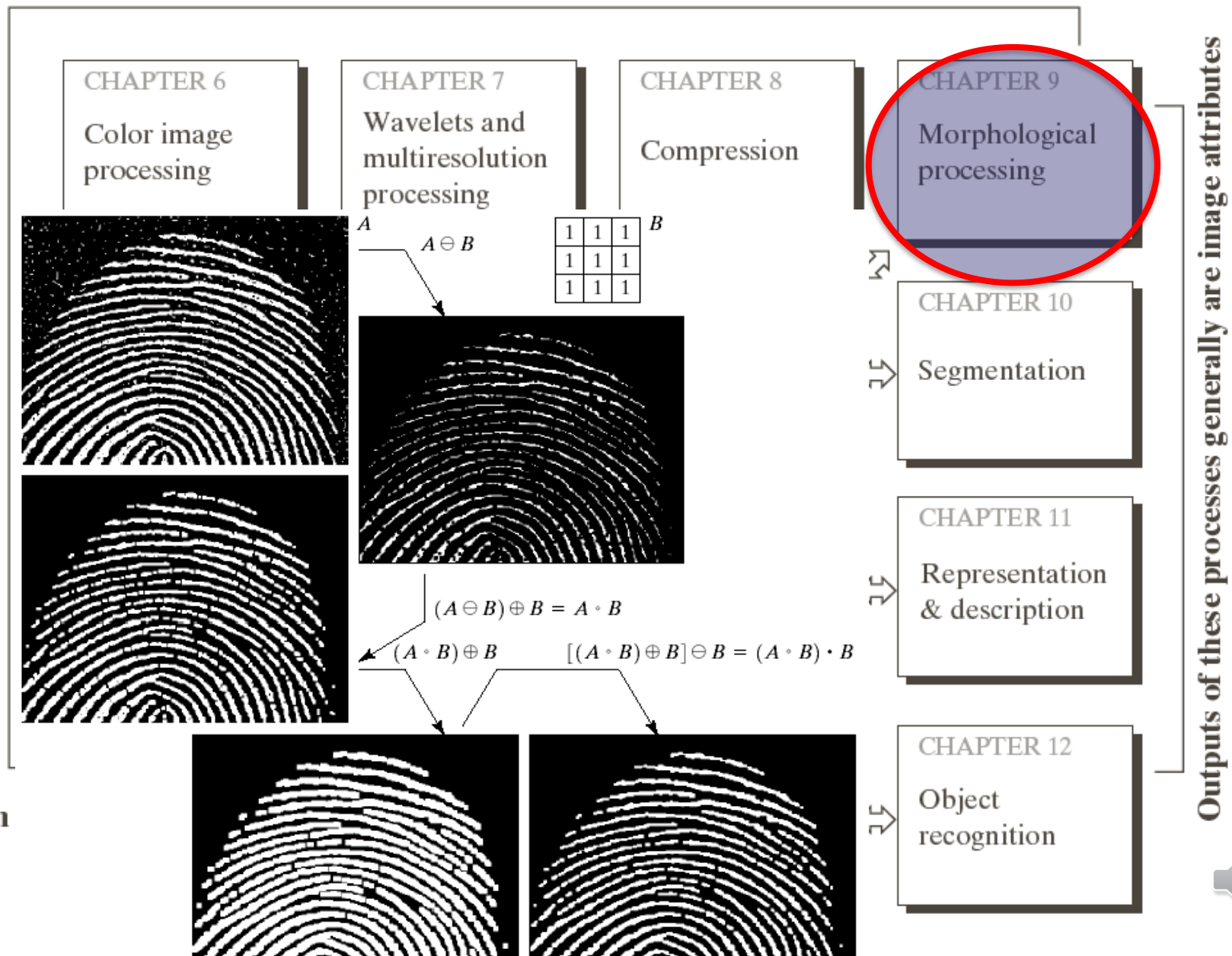
Outputs of these processes generally are image attributes



Outputs of these processes generally are images



Outputs of these processes generally are images





Outputs of these processes generally are images

CHAPTER 6

Color image  
processing

CHAPTER 7

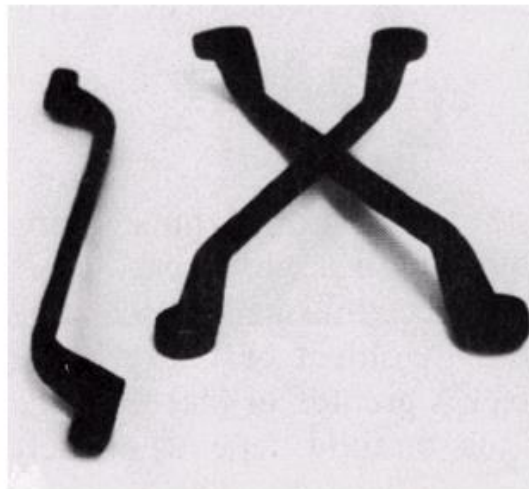
Wavelets and  
multiresolution

CHAPTER 8

Compression

CHAPTER 9

Morphological  
processing



CHAPTER 10

Segmentation

CHAPTER 11

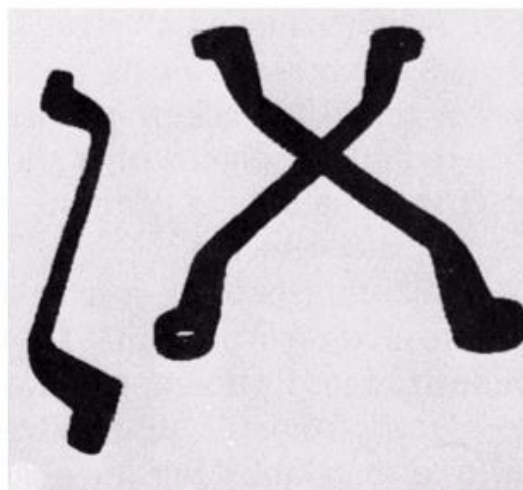
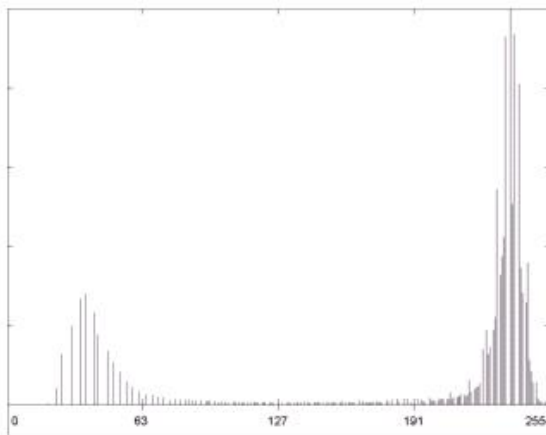
Representation  
& description

CHAPTER 12

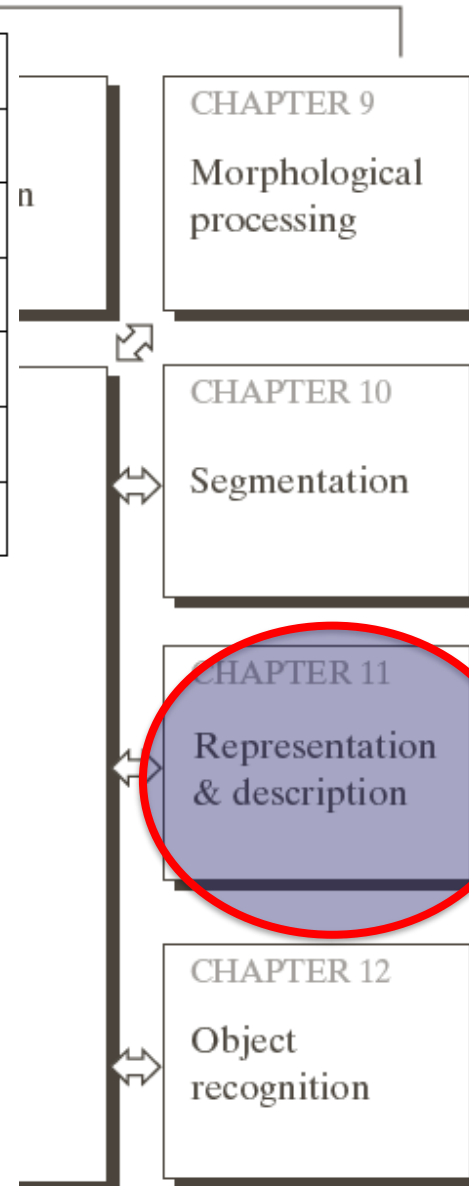
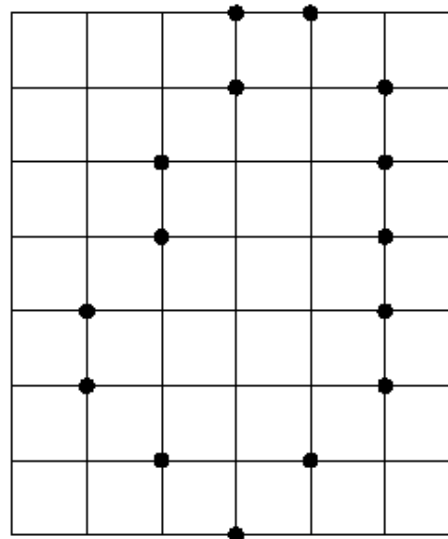
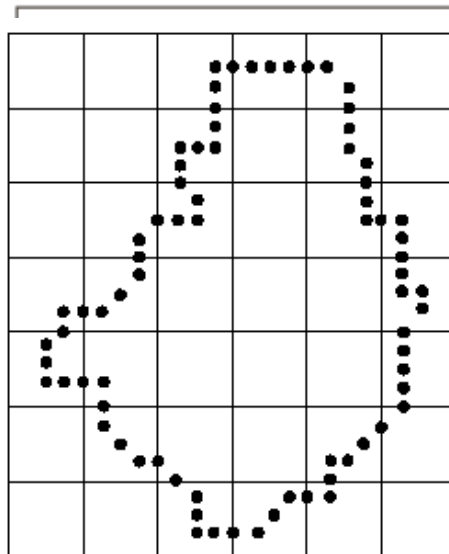
Object  
recognition

Outputs of these processes generally are image attributes

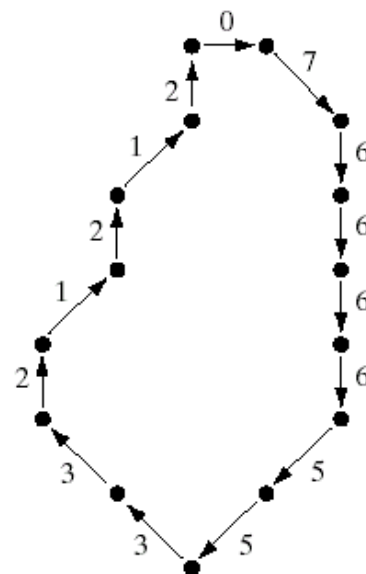
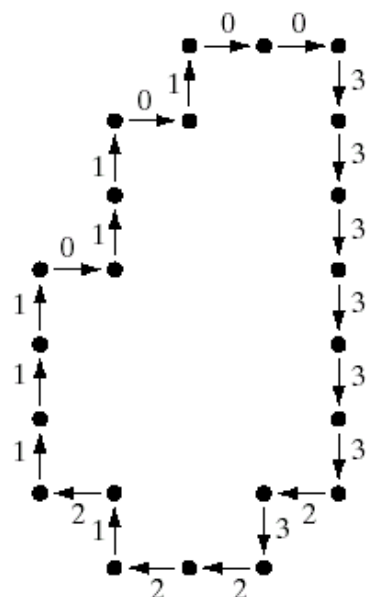
Problem  
domain



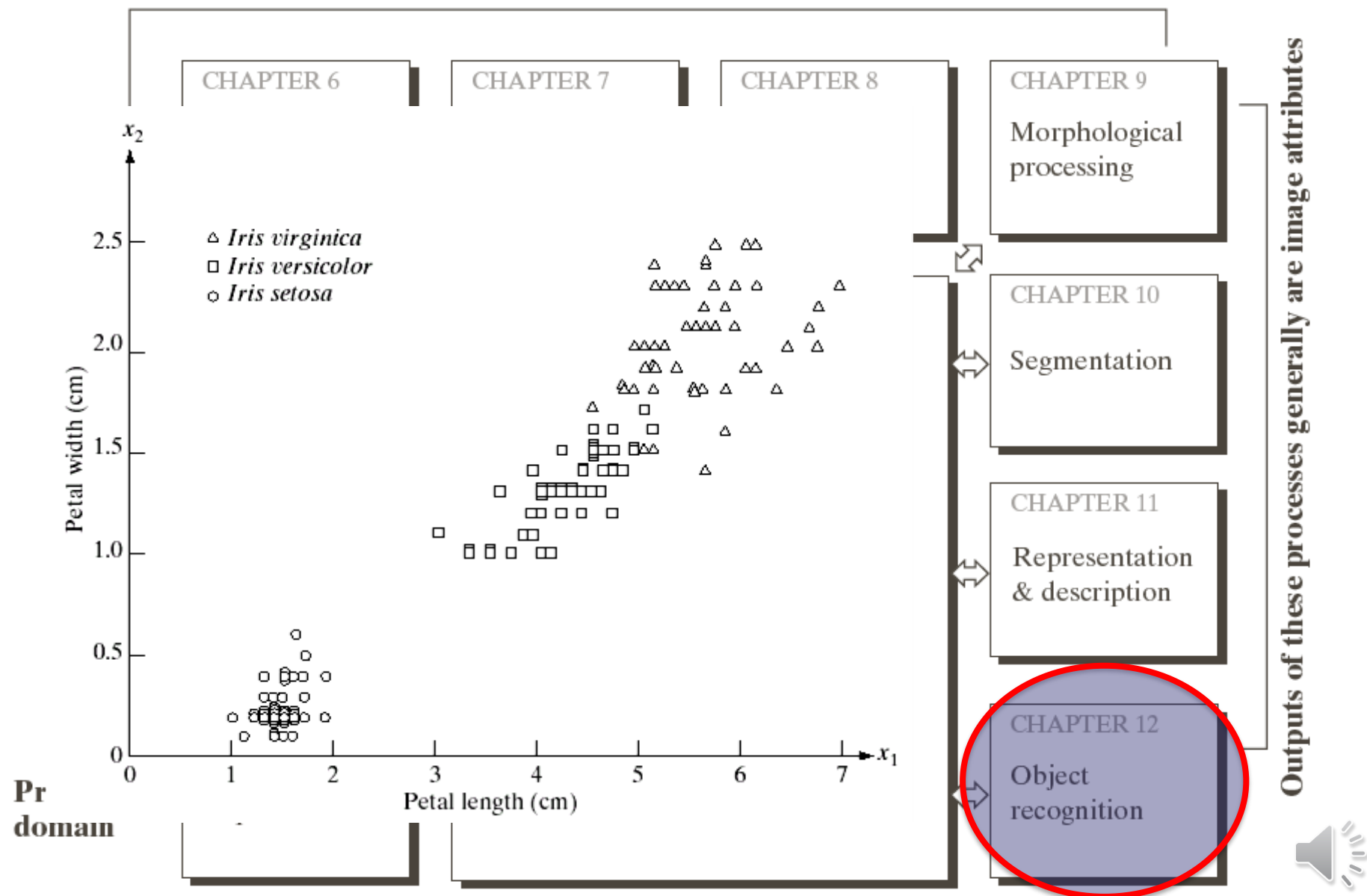
Outputs of these processes generally are images



Problem domain



Outputs of these processes generally are images



## Summary

We have looked at:

- What is a digital image?
- What is digital image processing?
- Why do digital image processing?
- History of digital image processing
- State of the art examples of digital image processing
- Key stages in digital image processing

Next time ...





## Assignments

- Find an example of digital image system, and explain every part?
- Give an example of digital image processing in one field?



# Questions?

